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Feasibility Boundaries for Authenticated UAV Telemetry

An Exclusion Bound and a Capacity Envelope

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Abstract

Flying ad-hoc networks (FANETs) stream telemetry records so small that a cryptographic signature (64–96 B) exceeds the record it signs: compactly encoded, a record is 24.0 B, and 9.0 B as a difference from its predecessor. Once the payload has been compressed, authentication — not data — becomes the dominant on-air cost, and the operational question stops being *how many bytes does a scheme save* and becomes *which configurations can run at all*.

This thesis answers the second question. It searches four design axes jointly — record *encoding*, authentication *placement*, signature *scheme* and *batch* size — against a byte model, an NS-3-validated IEEE 802.11 broadcast channel model, and energy measured on Raspberry Pi 4 hardware. Three results follow, ordered by how durable they are.

An exclusion. A frame that verifies alone must carry a header, a chain link, a signature and one record that decodes without its predecessor. A link whose payload cannot hold those admits no such telemetry at any batch size, and packet loss forecloses fragmentation as an escape whenever the verifiability target is no looser than the loss rate. On LoRaWAN this removes eight of the twelve EU863-870 data rates: five because a 64 B signature alone overflows a 51 B payload, and three because header, link and signature fill a 115 B payload before any record. The conditions under which that holds, and what each relaxation buys, are stated with it.

Bytes. Built as one frame format with a sender and a receiver, the design puts 43.25 B per record on air where per-record signing puts 145.77 B (70.3% fewer), at an operating point compliant with 3GPP TS 22.125. Every frame carries one self-contained record: a frame that depends on its predecessor does not meet the verifiability target at its own loss rate.

Capacity. Simulated directly, with a bootstrap interval, one 802.11a collision domain delivers 95% of frames from 124 neighbours with the design and from 35 with per-record signing.

In the byte model the axes were first searched with, only placement and batching interact, by exactly $g_a(1-1/b)$; in the frame as built, delta coding is conditional on batching too, since a frame that must decode alone starts with a full record. The signature-scheme axis is byte-neutral among 64 B schemes. AUTHBC introduces no new cryptographic primi-

tive. Its contribution is a reproducible co-design, its quantified trade-offs, the boundary beyond which no batch size helps, with the conditions under which it holds and what each relaxation buys — and a methodological record of the defects found while establishing it, including three results this thesis had to correct after an external review.

Declaration

► *TO BE COMPLETED by the candidate. Institutional wording required.*

Use of generative AI.

► *The candidate must state this explicitly and accurately; the repository history records the extent of agent involvement in implementation, experiment execution and drafting. Institutional policy governs the required wording.*

Acknowledgements

► *TO BE COMPLETED.*

Nomenclature

Symbols follow docs/01_SYSTEM_MODEL_ARCHITECTURE.md §2, which is the single source of truth for notation across the repository, the papers and this thesis.

Symbol	Meaning	Default
s	encoded record payload bytes	measured (E1)
g_a	authentication object bytes (scheme σ)	64 / 64 / 96
H_f	ledger frame header bytes	44 first format, 23 lean; measured
L	chain link carried once per frame (lean format)	35
s_k, s_δ	self-contained record; record coded against its predecessor	measured
r	frames per self-contained frame (reference interval)	1
M	application MTU budget	1500 (802.11)
b	records per frame (batch)	decision variable
p	frame loss probability	{0.02, 0.05, 0.10}
ε	verifiability slack; require $V \geq 1 - \varepsilon$	$= p$
Λ_i	per-UAV record rate (rec/s)	50 (adopted)
Λ	aggregate rate a receiver verifies	$\Lambda_i \cdot N_{\text{local}}$
N_{local}	neighbours in the collision domain	reported as a curve
U	channel utilisation, offered / deliverable	admission metric
V	$P(\text{record verifiable at receiver})$	constraint $\geq 1 - \varepsilon$
R	PHY data rate	6 Mbit s ⁻¹
$D(b)$	freshness of the oldest record in a batch	$\leq D_{\text{max}}$
s_{max}	largest record a single frame admits	$M - H_f - L - g_a$
N_{max}	largest sustainable neighbourhood	derived

A note on H_f . It is reported as a *range*, not a constant. Canonical CBOR encodes small integers in one byte and large ones in five, so the measured frame header moves by six bytes with the magnitudes of the sender identifier and sequence number (Table 3.3).

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Chapter 1

Introduction

1.1 The problem, in one paragraph

A flying ad-hoc network of unmanned aerial vehicles must do two things at once that pull against each other. It must broadcast telemetry often enough to be useful — position, velocity, battery, mode, tens of times a second — and it must make that telemetry *tamper-evident*, so that a record cannot be forged, replayed or silently altered by a participant or an outsider. The second requirement costs bytes. A digital signature at the 128-bit security level is 64 B for Ed25519 or ECDSA-P256, and 96 B for the BLS12-381 variant actually shipped by the reference implementation. A telemetry record, once compressed, is 45 B — and 32 of those are not telemetry either: they are the hash that links the record to its predecessor. The data are 13 B.

That inequality is the whole thesis. **The signature is larger than the thing it signs.**

1.2 Why this is not simply a compression problem

The obvious response — compress harder — fails, and it fails for a reason worth stating precisely. Payload compression was the subject of prior work; this thesis begins where that work ends. Once a record is squeezed from 191 B of JSON to 45 B of delta-encoded CBOR, the signature has gone from 25% of the on-air cost to 59%; counted with the chain link inside the record, signature and link are 96 of 109 B. Every further byte saved on the payload increases the fraction spent on cryptography. Compression does not solve the problem; *compression is what creates it*.

The design space that remains has four axes, and this thesis searches them jointly:

- **Encoding** e — how a record is serialised (JSON, CBOR, MessagePack, delta-CBOR).

- **Placement** — *where* the signature sits relative to the records it covers: inline per record, self-batched over b of one sender’s records, cross-signer aggregated, or block-level across multiple frames.
- **Scheme** σ — which signature algorithm, and therefore how many bytes the authentication object costs and how much CPU verification demands.
- **Batch** b — how many records share one authentication object, bounded by the frame size on one side and by a freshness deadline on the other.

1.3 The question this thesis actually asks

The literature on efficient authentication asks *how many bytes does my scheme save*. That is a useful question and it has been answered many times. This thesis asks a different one, because for the deployments that motivate it the byte count is not the binding constraint:

Which configurations can run at all?

That reframing — from optimisation to feasibility — is the thesis’s organising decision, and Chapter 11 records why it was made partway through the work rather than at the start. Briefly: as a co-design optimisation the result is mid-tier, because the optimisation is largely closed-form and a factorial ablation showed only one genuine interaction among the four axes. As a *boundary* result the same evidence is stronger, because an impossibility cannot drift as models improve — and four of this project’s own performance numbers did drift, and had to be corrected.

1.4 Research questions

- RQ1** How large is authentication overhead after compression, and where is the payload-size threshold below which it dominates?
- RQ2** Which authentication placement minimises on-air bytes while keeping records verifiable under loss, and what does self-batching change relative to cross-signer aggregation?
- RQ3** When does an aggregatable scheme beat a fast conventional one, as a function of relay fraction, arrival rate and the radio/CPU power ratio?
- RQ4** What is the end-to-end optimal configuration, how much does it beat the naive and compression-only baselines, and do the analytical models survive contention simulation and real hardware?

RQ5 (*added during the work*) Where is the boundary beyond which a signed, hash-chained frame cannot be sent at all, whatever the batch — under which conditions does it hold, and what does each relaxation (a shorter authenticator, no chain link on air, loss recovery) buy?

RQ5 was not in the original charter. It emerged from the low-rate arm and became the thesis’s most durable result; Chapter 10 develops it, and records the two occasions on which this thesis stated it wrongly.

1.5 Contributions

1. **An overhead-dominance characterisation** for compressed authenticated FANET ledgers, with the threshold stated in closed form and measured across four encodings (Chapter 4, Chapter 7).
2. **A placement theory** distinguishing self-batching from cross-signer aggregation, with a loss-robustness frontier proving that frame-level batching Pareto-dominates block-level under any stationary loss process — not merely under the independent-loss assumption the emulator implements (Chapter 4).
3. **An exclusion bound instantiated on a real band, with its scope.** Eight of the twelve EU863-870 data rates cannot carry one signed, hash-chained frame: five because a 64B signature will not fit a 51B payload, three because header, chain link and signature fill a 115B payload before any record. The four conditions under which that holds, and what each relaxation buys, are stated with it (Chapter 10).
4. **The design built as one object, and its capacity measured directly.** A frame format with a sender and a receiver, in which every reported size is the length of an emitted frame; and the largest neighbourhood each configuration sustains, found by simulating each at its own frame size and rate with a bootstrap interval: 124 neighbours for the design against 35 for per-record signing (Chapter 3, Chapter 8).
5. **Validated models with their gaps stated,** including a broadcast channel model validated to $\leq 0.08\%$ against an independent slot-exact simulator and $\leq 0.51\%$ against NS-3, plus two-radio hardware measurement of the airtime constants (Chapter 9).
6. **An ablation that narrows our own claim,** showing that in the byte model only placement and batching interact, that an apparent encoding interaction was a ratio-scale artifact, and that the one real coupling of encoding to batching appears only when the frame is built (Chapter 8).
7. **A reproducible testbed and a defect taxonomy.** Every model-derived artifact is re-derived byte-identically by an automated gate, and the classes of defect

that survived that gate are catalogued — including the ones found in this work (Chapter 11).

1.6 Explicit non-goals

This thesis introduces no new cryptographic primitive, no new consensus protocol and no post-quantum implementation; post-quantum sizes appear only as analytical projections. Key compromise, multi-hop routing research and finality semantics are out of scope. The low-rate arm is a *modelling* chapter: no LoRa radio was metered and no energy column exists for it.

1.7 Structure of this thesis

Chapter 2 places the work in its literature. Chapter 3 fixes the system and threat models and the notation. Chapter 4 develops the theory. Chapter 5 describes what was built and Chapter 6 how it was measured. Chapter 7 and Chapter 8 give the 802.11 results; Chapter 9 the channel-model validation and hardware; Chapter 10 the low-rate regime and the exclusion bound. Chapter 11 is unusual for a thesis and deliberate: it reports the errors found in this project’s own results, how they were found, and what class of defect resisted the safeguards. Chapter 12 concludes.

Chapter 2

Background and Related Work

► *STATUS: OUTLINED, PARTLY DRAFTED. The positioning below is adapted from the paper's related work. A thesis requires materially broader coverage than a paper: each section below names the reading that must still be done. Do not submit this chapter as it stands. See `thesis/STATUS.md`.*

2.1 How citations are checked in this thesis

No figure or number is quoted from a source that has not been read in full, and where a source establishes prior art for something this thesis might otherwise have claimed as novel, that is said explicitly. Every bibliography entry is compared by a script with its Crossref, DataCite or arXiv record — family names in order, initials, title and year — and an entry with no registry record must name the held document it was checked against.

Those rules exist because each was broken once. An early finding claimed this work was the *more optimistic* of two LoRa capacity models; it was drawn from the wrong figure of a paper that reported pure-ALOHA results beside LoRa results, and was retracted: **quoting the paper is not enough — quote the figure**. And until October 2026 this section said that every reference had been confirmed against its Crossref record, while four entries carried wrong metadata and one paper was cited under the authors of a different one. The confirmation had checked that each DOI resolved; it had not compared the author list. **A claim that citations were checked belongs in a check that runs**, which is why the sentence is gone and the script exists (Chapter 11).

2.2 Flying ad-hoc networks and UAV telemetry

► *TO WRITE. Required coverage: FANET as distinct from MANET/VANET (mobility degree, topology change rate, energy asymmetry); the telemetry workload itself, anchored to autopilot source rather than secondary summaries — PX4's MAVLINK_MODE_ONBOARD at 50 Hz and ArduPilot's vehicle-specific defaults; and 3GPP TS 22.125 S5.2.2 as the independent standards anchor for*

direct UAV-to-UAV broadcast. Sources are held; the synthesis is not written.

2.3 Tamper-evident telemetry and hash-chained ledgers

Linking each record to the hash of its predecessor makes an append-only log tamper-evident: altering a record invalidates every subsequent link. The construction is Haber and Stornetta’s [1]; Bitcoin made it the spine of a blockchain [2]. UAV-blockchain systems, surveyed in [3], place telemetry or attestations on such a chain, gaining integrity at a per-record overhead cost. This thesis targets exactly that substrate cost and is agnostic to the consensus layer above it; finality and witnessing semantics are out of scope.

► *TO EXPAND: the distinction between a hash chain (integrity of ordering) and a signature (attribution) is load-bearing for the threat model in Chapter 3 and deserves a full subsection.*

2.4 Signatures for constrained authentication

ECDSA [4] yields 64 B P-256 signatures at 128-bit security but is randomness-sensitive. Ed25519 [5, 6] uses a twisted-Edwards curve with deterministic nonces for fast, misuse-resistant, batchable verification. BLS [7] gives the shortest signatures via pairings, and [8] aggregates n signatures into one at the cost of expensive pairing verification.

AUTHBC introduces *no new primitive*. It quantifies *where and when* each scheme wins once placement and batching are fixed (Chapter 8), finding Ed25519 dominant for self-batching and BLS worthwhile only for cross-signer relay traffic.

Independent work on UAV swarms adopts precisely the design this thesis treats as its naive baseline: Rotta and Mykytyn attach a HMAC-256 signature with a timestamp to *each* broadcast telemetry message, and observe that MAVLink telemetry messages are much smaller than 256 bytes [9] — corroborating both the premise and the baseline from outside this work. Their construction is a symmetric MAC under a group key, so it provides integrity and group authenticity but *not* non-repudiation, and therefore cannot underwrite a ledger attributable to a specific UAV. That is a different trust model, not a cheaper solution to this one.

2.5 Authenticating messages one at a time, and signing streams

2.5.1 Per-message practice

MAVLink 2 signing appends a link identifier, a 48-bit timestamp and a 48-bit truncated SHA-256 tag computed with a shared secret [10]: thirteen bytes per message. It is cheap

and it authenticates a *link* — any holder of the key can produce any message — so a record cannot be attributed to one vehicle afterwards. Vehicular broadcast uses public-key signatures: an ECDSA signature on each message, with the signer’s certificate in some messages and an eight-octet digest of it in the others [11, 12]. That is the trust model adopted here, and the per-message cost this thesis starts from.

2.5.2 Amortising one signature over many packets

This is a classical problem and the closest prior art to batching. Gennaro and Rohatgi sign the first packet of a stream and chain the rest by hashes [13]; one lost packet breaks the chain. Wong and Lam sign a hash tree over a block and send in every packet the hashes needed to reach the root [14], so each packet verifies alone — the *signing cost* is amortised, the bytes are not. EMSS tolerates loss by carrying several hashes per packet with a periodic signature packet, and TESLA replaces the signature by a MAC whose key is disclosed later [15, 16]; TBRD applies TESLA to UAS Remote ID and reports about half the overhead of signatures [17].

All of these spread *one authenticator across packets*, and so make packets depend on each other or on a later one. This thesis does the opposite: it puts *several records in one packet*, which is possible only because a telemetry record is a small fraction of a frame. Each frame then stays self-verifying and the cost moves to latency, at the sender. TESLA instead delays verification at the receiver, and a disclosed MAC key gives no non-repudiation, which a ledger needs. The lesson of the stream-signing literature that does transfer is the one this thesis learned late: a unit that depends on its predecessor inherits its predecessor’s loss (Theorem 4.6). Section 8.4 places each of these schemes at this thesis’s operating point, by authenticator bytes and by frames per second.

2.5.3 Hash-linked records

Linking each record to the hash of its predecessor to make a sequence tamper-evident is due to Haber and Stornetta [1] and is the chain in every blockchain. The LoRaWAN link layer whose payload limits bound Chapter 10 is specified in [18, 19].

► *TO EXPAND: implicit (ECQV) certificates from IEEE 1609.2, never priced here; and the simulation-credibility lineage [20], which Chapter 11 should be read against.*

2.6 Batch and aggregate verification

Zhang *et al.* [21] let a roadside unit verify multiple received signatures simultaneously, motivated by the same pressure from the opposite side: a hard deadline — DSRC’s 300 ms — within which sequential verification does not scale. Their batching is *receiver-side* and buys verification time; the batching in this thesis is *sender-side* and buys airtime. The

two are complementary rather than competing. The nearest FANET work is of the same kind: Rajasekaran *et al.* [22] authenticate a group of drones to a user in one batch check of a pairing-based scheme, and report the communication cost of n authentications as n times that of one (1184 n bits) — the verifier’s work is shared, the bytes are not. This work generalises from receiver batching to authentication *placement*, co-designs it with encoding and scheme, and adds a loss-robustness frontier that pure aggregation does not address.

Certificateless aggregate signature (CLAS) schemes for vehicular broadcast are the closest external comparator. Li *et al.* [23] tabulate the per-message size of their own scheme and four others, for a 67 B traffic message: 583 B (theirs, and Cahyadi *et al.*), 596 B (Xu *et al.*), 735 B (Liang *et al.*) and 859 B (Wang *et al.*) — the last four as reported there, not read at source. In every one the cost of n messages is n times the cost of one. **The finding is the axis, not a ratio:** aggregation compresses the verifier’s work, while each message still carries its own identity, key and signature components; batching compresses what is sent. No claim is made to beat those schemes. They provide conditional privacy, which this work does not, their group elements are 128 B against a 64 B signature here, and their figures are for a different message.

► *TO WRITE: a fuller treatment of CLAS constructions and their security models.*

2.7 Compact serialisation and delta coding

CBOR [24] and MessagePack [25] give compact binary encodings. Delta coding of telemetry time series is established practice, most prominently in Gorilla [26], which stores delta-of-delta timestamps and XOR-folded values for roughly a 10× reduction.

The variant used here differs in one respect the link forces. Gorilla is a database and can chain deltas indefinitely; a lossy broadcast link makes an unbounded chain fragile, so this work re-anchors on a keyframe every K records and pays the resulting size for loss independence.

2.8 Channel models for IEEE 802.11

Bianchi’s fixed-point analysis of the DCF [27] is the standard model for saturated unicast throughput. Broadcast is *not* that model with the acknowledgement removed — a point Ma and Chen make explicitly, and which this project rediscovered expensively. Their broadcast model [28, 29] accounts for the backoff-counter Consecutive Freeze Process, which dominates when the contention window is small relative to the station count. Tinnirello, Bianchi and Xiao [30] supply the anomalous-slot refinement that explains a residual small-frame bias observed in Chapter 9.

2.9 LoRa and LoRaWAN capacity

Haxhibeqiri *et al.* [31] provide the only published LoRa capacity model found that is grounded in hardware interference measurements rather than an analytical collision assumption, and state it in closed form so it can be evaluated at this work’s operating point rather than merely quoted at theirs. Zirak *et al.* [32] supply the only air-to-air LoRa PDR-versus-range field data located, which range-limits the capacity result of Chapter 10.

► *TO EXPAND: LoRaWAN’s regional parameters as a regulatory rather than a technical constraint — the EU868 duty cycle is enforced per sub-band, which is why adding channels buys no extra airtime. This is developed in Chapter 10 but belongs partly here.*

2.10 Where this thesis sits

The adjacent literature almost uniformly proposes a *new construction* and evaluates it against its predecessors. This thesis deliberately proposes none. Its contribution is a boundary: the links on which a frame that verifies alone cannot be sent at any batch size, stated with the conditions under which that holds. For a given authenticator the result is arithmetic rather than empirical, so unlike a performance figure it does not move as models or hardware improve. It does move with the authenticator — a 48 B signature or a 13 B tag fits where a 64 B signature does not — and Chapter 10 gives the matrix. It is invisible to the byte models the literature optimises against.

► *The bibliography holds every source that is held and has been read, and stops there deliberately. A thesis of this scope would normally carry more; reaching a higher count requires a genuine sourcing pass — find, download, read, then cite. Padding the list would be the same defect this project spent its audit removing.*

Chapter 3

System Model and Threat Model

► *STATUS: DRAFTED from docs/01; revised October 2026 (two frame formats, a full threat model, capacities from direct simulation). Still needs one worked example of a frame on the wire, byte by byte.*

3.1 Network

N UAVs share an IEEE 802.11 broadcast channel (OFDM, base rate $R = 6 \text{ Mbit s}^{-1}$ for robustness; application MTU budget $M = 1500 \text{ B}$). Frames are lost independently with probability p , an abstraction of fading, collisions and mobility, with $p \in \{0.02, 0.05, 0.10\}$ as a declared sensitivity range. Contention is modelled by the DCF (Chapter 4) and validated against NS-3 (Chapter 9). There is no ground control station in the model: the substrate is peer-to-peer broadcast.

3.1.1 Why broadcast, and why it matters

IEEE 802.11 broadcast frames carry no acknowledgement and are never retransmitted. Three consequences follow, and each shapes a later chapter. A broadcast receiver sees the *raw* channel error rate with none of the ARQ that hides loss on a unicast link, which is why the loss grid is one to two orders of magnitude more pessimistic than the 99.9% reliability TS 22.125 specifies for managed command-and-control links *with* retransmission. Overload degrades *delivery*, never latency, because excess offered load is dropped rather than queued. And the channel model is not Bianchi's — Chapter 9 records that using the unicast model with the acknowledgement removed under-predicts simulation by $16\times$.

Table 3.1: The two operating points, lean format. Bytes are identical; the neighbourhood is not. $N_{\max}$ is the largest neighbourhood with mean delivery ≥ 0.95 , found by direct simulation (Chapter 8).

	Λ_i	$D_{\max}$	b	B/rec	$N_{\max}$, per-record signing	$N_{\max}$, design	TS 22.125
adopted	50	100 ms	4	43.25	35	124	yes
alternative	20	250 ms	4	43.25	85	306	no

3.2 Ledger

Each UAV maintains an append-only hash chain of its own telemetry records and stores verified records overheard from others. A record is

$$\text{rec} = \{\text{src}, \text{seq}, \text{ts}, \text{prev_hash}, \text{payload}\}, \quad \text{prev_hash} = \text{SHA-256}(\text{previous record bytes}).$$

Only the owner appends to its own chain — an owned-object model — so tampering or equivocation on a (src, seq) pair is detectable by chain verification. Finality and witnessing semantics are out of scope; the deliverable is *authenticated dissemination*, not consensus.

3.3 Traffic and the two arrival rates

Records are generated at rate Λ_i per UAV. The aggregate neighbourhood arrival rate is $\Lambda = \Lambda_i \cdot N_{\text{local}}$ — the rate a *receiver* must verify.

Remark. These are different quantities and conflating them is a real defect, recorded as such. A node batches its *own* records, so freshness uses Λ_i ; it verifies *everyone's*, so the verification-throughput constraint uses Λ . Using Λ_i for both silently models a one-sender network and makes cross-signer aggregation look feasible at fleet sizes where the receiver cannot keep up.

3.3.1 The adopted operating point

$\Lambda_i = 50 \text{ rec/s}$ at $D_{\max} = 100 \text{ ms}$ is adopted. It is PX4's MAVLINK_MODE_ONBOARD companion-computer rate, read from autopilot source rather than from a secondary summary, and it is fully compliant with 3GPP TS 22.125 S5.2.2, which specifies at least 10 messages per second and at most 100 ms end-to-end latency for direct UAV-to-UAV local broadcast.

The alternative (20 rec/s, 250 ms) point is retained and reported because it gives *identical bytes* — batching obeys $b \leq \Lambda_i D_{\max}$, so only the product matters, and both give $\Lambda_i D_{\max} = 5$, hence $b = 4$ and the same frame.

3.3.2 The floor of the region, which is a real constraint on the application

Holding $b \geq 4$ under a 100 ms deadline requires $\Lambda_i > 40$ Hz. At 20 or 10 Hz the same deadline forces $b = 1$ and nothing is left of the saving but the choice of encoding. **The batching benefit is not available at every compliant operating point:** it requires a telemetry rate fast enough to fill a batch inside the deadline. More structurally, any batching at all needs $\Lambda_i \geq 1/D_{\max}$ — at the standard’s 100 ms bound, exactly the standard’s own minimum message rate. At an ArduPilot Copter default of 0 Hz, where streams are requested by a ground station rather than pushed, this thesis’s mechanism does not apply at all.

3.4 Threat model

3.4.1 The adversary

The adversary controls the radio. It can read, inject, replay, modify, delay and drop frames, and may operate any number of radios of its own. It does not hold any honest UAV’s private key; key compromise is out of scope by charter.

3.4.2 What is claimed, and where each property is enforced

property	meaning	mechanism
origin authentication	a stored record was produced by the UAV named in <code>src</code>	public-key signature over the canonical records, checked before anything is stored
integrity	no field of a stored record was altered in transit	the same signature: every one of 1 392 single-bit flips of a design frame is refused
replay rejection	a frame delivered twice is stored once	<code>seq</code> strictly increasing per sender
equivocation evidence	two validly signed records for one (<code>src</code> , <code>seq</code>) are kept as a pair	both carry the sender’s signature, so the pair is transferable proof
continuity	omission or re-ordering within what is held is detectable	each record’s <code>prev_hash</code> ; one link per frame commits to the record before it
non-repudiation	a sender cannot later deny a record	follows from a public-key signature; a symmetric tag would not give it

3.4.3 What is not claimed

Each of these is outside the design, not an oversight in it.

Confidentiality. Telemetry is broadcast in clear.

Availability. Jamming removes frames, and nothing here recovers them: there is no retransmission and no coding across frames. Loss appears as the *robustness* measure V , not as a security property. A flood of forged frames costs a receiver one verification each, which the CPU budget of Chapter 8 bounds and does not prevent.

Timeliness against a delaying attacker. Replay is rejected by sequence number, not by time. An attacker who withholds a frame from a receiver and delivers it later gets it accepted — authentic, in order, and stale. Refusing that needs a clock shared with the sender, which is not assumed. The record’s signed timestamp lets an application apply its own bound.

Truthfulness. A UAV holding a valid key can sign false telemetry. Only *inconsistent* falsehood — two records for one sequence number — is detectable.

Key distribution, revocation and admission. Receivers are assumed to hold each sender’s public key. The on-air cost of delivering certificates is charged separately (Chapter 8); who may join and how a key is withdrawn is out of scope, and so is Sybil resistance.

Privacy. Senders carry a stable identifier. The certificateless aggregate schemes of the vehicular literature buy conditional privacy that this design does not offer.

3.4.4 Behaviour under loss

Every frame of the design decodes alone (Theorem 4.6), so a lost frame costs its own records and nothing else. The receiver then holds a chain with a gap: the next frame’s link commits to a record it does not have, which it records and cannot check. Continuity across the gap is therefore *provable later*, if the missing records are ever obtained, and *unknown until then* — neither violated nor verified.

Remark. Availability under loss is a *robustness* metric in this thesis, not a security claim. The quantity V , the probability that a record is verifiable at a receiver, is treated throughout as a design constraint, not as a guarantee against an adversary who controls the channel.

3.5 The decision space

3.5.1 Authentication placements

- A — inline per record.** Every record carries its own signature. The baseline.
- B — self-batch.** One UAV packs b of its *own* records into a frame and signs the frame once. Per-record authentication cost $(g_a + H_f)/b$. Self-contained, hence loss-local.
- C — cross-signer aggregate.** A frame carries b records from *different* originators, each originally signed; the signatures aggregate into one. Required only when forwarding others' records or combining attestations.
- D — block-level aggregate.** One signature over b records spanning $n > 1$ frames. All-or-nothing under loss; Chapter 4 shows when this is dominated.

3.5.2 Schemes and encodings

Signature schemes are ECDSA-P256 (64 B, legacy baseline), Ed25519 (64 B, fast) and BLS12-381. The BLS object is 96 B, not the 48 B minimum-signature variant originally specified: the available library ships only the minimum-pubkey scheme. That deviation is recorded and applied consistently, and it makes BLS lose the byte comparison outside multi-signer aggregation.

Encodings are JSON (baseline), canonical CBOR, MessagePack, and delta-CBOR with keyframes every $K = 16$ records. K is fixed and not optimised — a real design variable left on the table, and declared as such.

3.6 Two frame formats

The thesis reports two formats side by side, and the reason is historical. The *first* format is the one implemented at the start: canonical CBOR with text keys, and in every record its own chain link, sender identifier and sequence number. It is frozen, and every artifact of chapters 7–9 that predates October 2026 was produced with it. The *lean* format was built after an external review pointed out that the design the results headlined — delta records under one signature in a short header — had never existed as one object: it was a sum of sizes measured in three places (Chapter 11). The lean format is that design as code, with a sender and a receiver, and every size reported for it is the length of a frame that was emitted, decoded and verified.

Table 3.2: Byte budget of a frame, field by field, for a sender with a five-digit identifier one hour into a 50 Hz flight (smaller values earlier in a flight in parentheses).

field	content	first	lean
(map)	CBOR map header	1	1
v, t	version; frame type	6	4
src	sender identifier, 16 bit	7 (5)	4 (2)
base_seq	sequence number of the first record	14 (10)	6 (2)
n	records in the frame	3	2
recs	key and length of the records	6	3
auth	key and length of the signature	7	3
header H_f		44 (38)	23 (17)
chain link		32 per record	35 per frame
signature g_a (Ed25519, ECDSA)		64	64
record that decodes alone (keyframe)		59.9*	24.0
record coded against its predecessor, 50 ms apart		44.0*	9.0

* includes that record's own 32 B chain link.

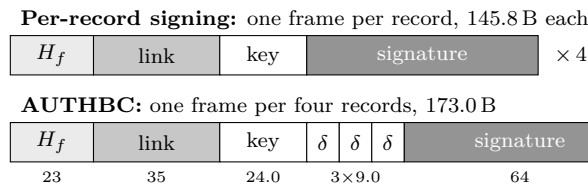


Figure 3.1: The lean frame, to scale (bytes below). Per-record signing repeats header, chain link and signature with every record; the design carries them once per four records, and the three records after the first are coded as differences (δ).

3.6.1 The lean frame

Figure 3.1 draws the frame to scale. A frame is one canonical CBOR map with integer keys. **src** and **base_seq** name the sender and the sequence number of the first record; **link** is the hash of the record before it; **recs** is the record stream and **auth** one signature. A record in the stream is nine variable-length integers — timestamp, position, velocity, battery, mode. The first record of a frame is a *keyframe* carrying absolute values; each later one is a *delta* from the record before it. Sender and sequence number are not repeated per record.

What is signed is not the stream. It is the canonical ledger records, each with its own **prev_hash**, exactly as in the first format. The receiver rebuilds them — the first **prev_hash** is **link**, each later one the hash of the record just rebuilt — and only then checks the signature. A frame that cannot be decoded therefore cannot be verified, and Theorem 4.6 counts that. The state against which a later frame would be decoded advances only after a frame has been verified *and* stored, so a replayed frame cannot desynchronise a stream.

Table 3.3: H_f is not a constant. Canonical CBOR encodes integers 0–23 in one byte and those ≥ 65536 in five, so the measured header moves with field magnitude.

src	base_seq	first format	lean format
0	0	38 B	17 B
24	24	40 B	19 B
256	256	42 B	21 B
40 000	180 000	44 B	23 B

What one link per frame gives up. In the first format every record carries its own `prev_hash`; in the lean format only the first record of a frame does, and the rest are derived. Inside a frame, order and content then rest on the frame signature alone, where the first format has a signature and an independent hash. A frame is signed and accepted as a unit, so no strength is lost — but one mechanism remains where there were two. The decision of July 2026 had scoped per-frame chaining to the low-rate arm; the lean format extends it to both, and that extension is recorded as a decision, not presented as a neutral encoding choice.

3.7 The frame header, which is a range

H_f feeds the batching theory, the exclusion bound, the batch bounds, the channel constraint and the energy model. It is measured from the implemented wire formats:

$$H_f = \text{len}(\text{frame}) - \text{len}(\text{records}) - \text{len}(\text{link field}) - \text{len}(\text{signature}),$$

the link term being zero in the first format, where the link lives inside each record.

The values 44 and 23 B are used throughout; they are the steady-state values for any realistic flight and the conservative end for every byte comparison.

Remark (A claim this section used to make). Earlier versions said the range decides the exclusion bound of Chapter 10: that one data rate was excluded only because the top of the range was used. That was withdrawn in October 2026. The bound it rested on charged neither the chain link nor a record that decodes alone, and with both charged the rate is excluded at every header size, including zero (Section 10.2.5).

Chapter 4

Theoretical Framework

► *STATUS: DRAFTED from docs/02. Proofs are complete for T1–T3 and T6; T5 is stated honestly as an empirical separability result rather than a theorem, which is what it is.*

This chapter develops the closed-form results. Each carries its empirical validation hook, and where a result is *applied* rather than novel, that is stated at the point of statement rather than buried in related work.

4.1 T1 — the overhead-dominance threshold

Theorem 4.1 (Overhead dominance). *With inline per-record authentication, the authentication byte fraction is $\varphi(s, g_a) = g_a/(s + g_a)$, and $\varphi \geq \alpha \iff s \leq g_a(1 - \alpha)/\alpha$. In particular authentication dominates ($\alpha = \frac{1}{2}$) exactly when $s \leq g_a$.*

Proof. $\varphi \geq \alpha \iff g_a \geq \alpha(s + g_a) \iff g_a(1 - \alpha) \geq \alpha s$. □

The threshold is trivial; its *instantiation* is the motivating result. At $g_a = 64$ B and measured record sizes, JSON at 191 B gives $\varphi = 25.1\%$, CBOR at 66 B gives 49.1%, and delta-CBOR at 45 B gives 58.7%. Compression moved the bottleneck from data to cryptography. Validated by E1 (Chapter 7), where a remark states what these record sizes contain: s includes the record’s own 32 B chain link, so φ understates the point.

4.2 T2 — frame-level batching and the amplification law

Placement B or C packs b records, one authentication object of g_a bytes and a header of H_f bytes into one frame, subject to $b \leq b_{\max}(s) = \lfloor (M - H_f - g_a)/s \rfloor$.

Proposition 4.2. *Per-record authentication overhead $\omega(b) = (g_a + H_f)/b$ is strictly decreasing, so the byte-minimal feasible batch is $b = b_{\max}(s)$. Relaxing integrality, per-record on-air bytes are $C(s) = s \cdot A$ with amplification $A = M/(M - H_f - g_a) \geq 1$.*

Proof. $\omega'(b) < 0$ gives the first claim. For the second, substitute $b = (M - H_f - g_a)/s$ into $s + \omega(b)$ and factor. $\square$

So every payload byte saved by compression saves A on-air bytes — *in the MTU-limited regime*. This is MTU-efficiency algebra, not a new law, and Theorem 4.3 shows the regime is never reached on 802.11.

4.3 T2a — which ceiling binds, and what compression is therefore worth

Theorem 4.3 (Binding constraint). *Let $b^* = \min(b_{\text{MTU}}, b_{\text{fresh}})$ with $b_{\text{MTU}} = \lfloor (M - H_f - g_a)/s \rfloor$ and $b_{\text{fresh}} = \lfloor \Lambda_i D_{\text{max}} \rfloor$. Then*

$$\frac{dC}{ds} = \begin{cases} A = \frac{M}{M - H_f - g_a}, & \text{MTU-limited,} \\ 1 \text{ exactly,} & \text{freshness-limited.} \end{cases}$$

Freshness binds exactly when $s < (M - H_f - g_a)/(\lfloor \Lambda_i D_{\text{max}} \rfloor + 1)$.

Proof. Differentiate C in each regime; in the second, b is constant in s . $\square$

The consequence is sharp and negative for the 802.11 arm. At $M = 1500$, $H_f = 44$, $g_a = 64$, $\Lambda_i D_{\text{max}} = 5$, the boundary is $s < 232.7$ B. Every encoding studied lies below it — delta 45, MessagePack 65, CBOR 66, JSON 191 — so **freshness binds for all of them and A is never operative on 802.11**. Compression pays exactly $1\times$, and the residual authentication cost is a floor $(g_a + H_f)/(\Lambda_i D_{\text{max}})$ that compression cannot touch.

On a low-rate link ($M = 242$) the boundary falls to $s < 29.7$ B, so the MTU binds and A is operative. **The leverage that motivates compression is exclusive to the low-rate arm**, which strengthens rather than weakens that chapter's motivation.

4.4 T3 — the loss-robustness frontier

Frame-level placements are self-verifiable, so $V_B = 1 - p$ and a loss costs b records. Block-level placement D spans $n(b) = \lceil (bs + g_a + H_f)/M \rceil$ frames under one signature, so $V_D = (1 - p)^n$.

Theorem 4.4 (Fragmentation is foreclosed). *The constraint $V \geq 1 - \varepsilon$ admits at most $n_{\text{max}} = \lfloor \ln(1 - \varepsilon)/\ln(1 - p) \rfloor$ frames. If $\varepsilon \leq p$ then $n_{\text{max}} = 1$: block-level authentication is infeasible beyond a single frame.*

Proof. Solve $(1 - p)^n \geq 1 - \varepsilon$ for n . At $\varepsilon = p$ the right-hand side is $1 - p$ and $n \leq 1$. $\square$

Theorem 4.5 (Robustness to correlated loss). *For any stationary loss process of mean rate p , whatever its correlation structure,*

$$V_D(n) = P(\text{all } n \text{ frames arrive}) \leq P(\text{one given frame arrives}) = 1 - p = V_B,$$

with equality only at $n = 1$.

Proof. A joint probability cannot exceed a marginal. □

This matters more than it first appears. The $(1 - p)^n$ form assumes independent loss, which the emulator also implements — so agreement between measured and theoretical V is a *consistency check, not a validation*. Theorem 4.5 rescues the ordering without the assumption: independence is the *worst case* for D, and correlation narrows the gap but cannot invert it. Gilbert–Elliott simulation at matched mean $p = 0.05$ confirms it: $V_D(n=2)$ rises from 0.9025 (independent) to 0.9498 at a mean burst of 160 frames — asymptotically $1 - p$, never past it, and never reaching the feasibility threshold.

4.5 T3' — a frame coded against its predecessor is not self-verifiable

Theorem 4.4 rests on the premise that each frame verifies alone. **That premise fails for a delta encoding whose keyframes are rarer than its frames**, and the design this thesis first headlined was one: a keyframe every $K = 16$ records with $b = 4$, so that only one frame in four could be decoded without the frame before it. A receiver cannot verify what it cannot rebuild — the signature is over the records, not over the bytes on air — so a lost frame takes every later frame with it until the next self-contained one.

Theorem 4.6 (Verifiability with dependent frames). *Let one frame in r be self-contained. Frame j of a group ($j = 1, \dots, r$) verifies iff it and the $j - 1$ frames before it in the group all arrive, so under independent frame loss p*

$$V(r, p) = \frac{1}{r} \sum_{j=1}^r (1 - p)^j.$$

$V(1, p) = 1 - p$, and $V(r, p) < 1 - p$ for every $r > 1$.

Corollary 4.7. *If $\varepsilon \leq p$, the requirement $V \geq 1 - \varepsilon$ admits $r = 1$ only: every frame must decode alone.*

$V(4, 0.05) = 0.881$. The design as first published therefore missed $V \geq 0.95$ at the loss rate it was specified for. Its byte figure had amortised one keyframe over four frames; its verifiability figure had treated the four as independent; and the two had never been

evaluated on one object, because no such frame had been built (Chapter 11). Measured with the encoder and decoder in the loop, $r = 4$ verifies 0.876 of records (Chapter 7).

For bursty loss the same counting gives, for a two-state process with mean loss $\bar{p}$ and mean burst length B frames,

$$V = \frac{1 - \bar{p}}{r} \sum_{j=1}^r (1 - a)^{j-1}, \quad a = \frac{\bar{p}}{B(1 - \bar{p})},$$

which reduces to Theorem 4.6 for memoryless loss and is *larger* for longer bursts: losses that arrive together waste fewer good frames. It never exceeds $1 - \bar{p}$, with equality only at $r = 1$, so bursts soften the penalty of a dependent design and never remove it.

The corrected design uses $r = 1$. What a longer interval would buy is small — 3.5 B per record between $r = 1$ and $r = 16$ — and a cleaner link would be needed to afford it.

4.6 T4 — scheme selection

For a UAV’s *own* records under self-batching, one ordinary signature covers b records, so Ed25519 and BLS are near-equivalent in bytes while Ed25519’s verification is far cheaper than a pairing. Ed25519 wins self-batching outright.

For *relayed* records from distinct signers, Ed25519 must carry $b \cdot 64$ B while BLS aggregates. BLS saves $\Delta(b)$ bytes per record but costs Δt extra CPU. BLS is energy-optimal iff

$$P_r \cdot \frac{8\Delta(b)}{R} > P_c \cdot \Delta t,$$

and throughput-feasible iff $t_{\text{agg-verify}}(b)/b \leq 1/\Lambda$. At $R = 6 \text{ Mbits s}^{-1}$ the radio side is $\approx P_r \cdot 75 \mu\text{s}$ against a CPU side of $P_c \cdot (1\text{--}3) \text{ ms}$, so **Ed25519 wins on 802.11 and BLS’s regime is low-rate links and attestation compression** — a falsifiable, regime-dependent rule.

4.7 T5 — the co-design result, stated as what it is

Remark (Not a theorem). T5 is often written as a co-design theorem. It is not one. The joint optimum is *empirically separable*, verified by exhaustive search over the full grid of 4 encodings $\times$ 3 schemes $\times$ 4 placements $\times$ 32 batches = 1536 points. A non-separable optimum would have been found had one existed; none was. That is evidence, not proof, and the distinction is kept.

The recipe: pick the smallest deterministic encoding e^* , placement B for own traffic or C for relay, batch $b^* = \min(b_{\max}(s), \lfloor \Lambda_i D_{\max} \rfloor)$ — on 802.11 the second term binds — and scheme per T4.

A factorial ablation (Chapter 8) narrows this further, and against the project’s own

earlier claim: only placement and batching genuinely interact, by exactly $g_a(1 - 1/b)$, hence not at all at $b = 1$. Encoding is perfectly separable *in that byte model*, where a record's size does not depend on its neighbours, and the scheme axis is byte-degenerate. In the frame as built the delta saving is $(1 - 1/b)(k - \delta)$ per record for a keyframe of k and a delta of δ bytes — conditional on batching in the same way (Chapter 8).

4.8 T6 — the authentication-exclusion threshold

Remark (Prior art, checked before publication). The inequality below is *established*, not novel. Gündoğan *et al.* compute exactly $M - H_f - g_a = s_{\max}$ for 802.15.4/NDN, and the post-quantum literature reports NIST signatures as incompatible with 5G SIB1's 372 B limit. That a fragmented unit delivers with $(1 - p)^n$ is likewise standard in 6LoWPAN. T6 is therefore a *synthesis* of two known facts, presented as an applied threshold. What has not been found stated elsewhere is the *composition*: that the sliced-signature workaround the literature proposes is foreclosed whenever $\varepsilon \leq p$. This check was run *before* publication precisely because an earlier novelty claim in this project was not.

Theorem 4.8 (Exclusion). *A frame that verifies alone must carry a header H_f , one chain link L , an authentication object g_a and at least one record that decodes alone, s_k . A link of maximum payload M therefore admits authenticated, hash-chained telemetry iff*

$$M \geq H_f + L + g_a + s_k,$$

each term taken at its smallest, and no choice of batch size, encoding of later records or aggregation changes this: those decide only how the space above the floor is used.

Fragmentation does not escape it, by Theorem 4.4: at $\varepsilon \leq p$ the entire verifiability budget is spent on the first frame, so a signed unit may not span two.

Remark (The statement this replaces). Until October 2026 the theorem read $s_{\max}(M) = M - H_f - g_a \geq s_{\min}$, with $s_{\min}$ the smallest record. It charged neither the chain link, which per-frame chaining moves into the frame, nor the requirement of Theorem 4.6 that the record decode alone; $s_{\min}$ had been taken as a mean delta record less its link. The consequences, including a claim this thesis had to withdraw twice, are in Section 10.2.5.

4.8.1 Four tiers, because each names a different escape route

tier	condition	what could fix it
signature	$M < g_a$	a smaller authentication object <i>only</i>
header	$g_a \leq M < g_a + H_f$	a leaner frame header
chain link	$g_a + H_f \leq M < g_a + H_f + L$	taking the link off the air
record	$g_a + H_f + L \leq M < g_a + H_f + L + s_k$	a smaller self-contained record

The tier structure is what makes the bound *constructive*: Chapter 10 evaluates it for all twelve EU863-870 data rates and tabulates what each relaxation buys.

4.8.2 Scope

The theorem is conditional, and Chapter 10 states the four conditions in full: the payload limit is that of the link layer in use; every frame carries a public-key signature; there is no loss recovery; and $\varepsilon \leq p$.

4.9 A withdrawn theorem, kept visible

An earlier version of this framework contained a seventh result, T7, claiming that channel capacity excludes configurations at $U \geq 1$, where U is offered load over *saturation* throughput. **It was withdrawn on the day it was written**, by its own validation experiment.

Saturation throughput is what the channel carries when every node is backlogged; a lightly loaded channel collides far less and carries substantially more. Simulation at $N = 50$ shows 98.8% of frames still delivered at $U = 1.00$, with the verifiability crossing near $U \approx 2.4$. The claimed exclusion does not exist, and a “50% ceiling” derived from it was an artifact of the wrong threshold.

A second structural error compounded the first: the theorem assumed overload manifests as delay. It does not. Broadcast has no ARQ and no queue buildup, so excess load is dropped, not queued — mean delay at nine times saturation is still under 3 ms. **Any capacity argument here must be made in V , not in D .**

What survives is a cost rather than an exclusion: because $b \leq \lfloor \Lambda_i D_{\max} \rfloor$, tightening the freshness deadline shrinks the batch and raises the frame rate the medium must carry. Meeting the standard’s deadline costs swarm size — 124 neighbours where the relaxed point carries 306 — and *nothing* in bytes. That is the measured statement, and it is a better result than the theorem it replaces.

Remark (The lesson, recorded). T7 was written from a model quantity treated as a physical limit, and published into the documentation before the experiment that tested it existed — an experiment already scheduled. Running it first would have prevented the claim.

4.10 Freshness, and a convention that had to be derived

The freshness of the oldest record in a batch is modelled as $D(b) = b/\Lambda_i + T_{\text{air}} + W_q$ and enforced as a hard constraint.

Remark (Which quantity b/Λ_i is). With periodic arrivals, b/Λ_i is the *batch-window duration* — open to transmit — while the *age of the oldest record* at transmission is $(b-1)/\Lambda_i$, because it waits out the remaining $b-1$ gaps. Both were confirmed exactly by discrete-event simulation under deterministic and Poisson arrivals. The formulation above names the second and computes the first.

It is retained as the *worst case*, and it is exactly that: a record timestamped at its sample instant may describe state up to $1/\Lambda_i$ older, and $(b-1)/\Lambda_i + 1/\Lambda_i = b/\Lambda_i$. So it bounds end-to-end latency including sampling quantisation. The cost of the choice is stated: the tight reading admits $b = 5$ at both operating points (36.4 B per record, -75.0%) where the worst case admits $b = 4$ (43.25 B, -70.3%). **This thesis under-reports its own saving by about 4.7 percentage points.**

Chapter 5

Implementation

► *STATUS: DRAFTED from docs/05. Needs a module-dependency figure.*

5.1 Design rule: one home per fact

The implementation follows one architectural rule that is worth stating because it is what makes the reproducibility claims of Chapter 11 possible. **Every quantity has exactly one definition, and everything else refers to it.** There is one function that produces the canonical bytes a signature covers; one airtime implementation shared by the channel model, the energy model and the simulation comparison; one place where experiment parameters live.

Where that rule was broken, defects followed. A signature-size constant duplicated between two modules produced a stale result that survived a re-freeze. An airtime approximation that existed in two forms produced a 12% discrepancy on acknowledgement frames. And the largest instance was not a duplicated constant but a *missing object*: for three months the design the results headlined existed only as a sum of sizes measured in three modules, with no frame, no sender and no receiver behind it — so the byte model, the verifiability model and the exclusion test each evaluated a slightly different thing and none could be checked against the others (Chapter 11).

5.2 The library

encodings/ The e axis: JSON, canonical CBOR (RFC 8949 S4.2), MessagePack, and delta-CBOR with keyframes every $K = 16$. Delta encoders are *stateful*; a fresh instance per record emits all keyframes and silently inflates sizes — a trap that cost real time.

crypto/ The σ axis: Ed25519 and ECDSA-P256 (both 64 B), BLS12-381 (96 B). Known-answer tests from RFC 8032, NIST CAVP and the IETF BLS draft are vendored

and must pass before any timing work.

ledger/ The data structure being authenticated: the canonical record, the hash chain, persistence, and verification with replay and equivocation detection.

placement/ The placement axis. `wire.py` is the first frame format and the one place that decides *what bytes a signature covers* — the audit boundary; it is frozen. `framer.py` holds its header constant and the batch bounds. `wire_v2.py` is the lean format of Section 3.6: integer keys, one chain link per frame, a variable-length record stream, and the same signed bytes as the first format. `session_v2.py` is the protocol around it — a sender that emits one self-contained frame per reference interval, and a receiver that decodes, verifies, stores and reports what it did with each frame (accepted, desynchronised, malformed, unknown sender, bad signature, replay, equivocation, tampered).

models/ The analytical layer: the DCF fixed point with exact OFDM-symbol airtime, the published broadcast model, the energy and freshness models, the Pareto search, the scheme crossover, and the low-rate PHY with every constant transcribed from primary specifications. `frame.py` holds the *one* definition of a frame from which the byte, verifiability and exclusion results are all computed, and whose arithmetic is held equal to emitted frames by test. `phy.py` makes the OFDM timing a parameter.

sim/ A slot-exact backoff simulator written *before* the published broadcast model was located, retained as a cross-check that shares none of that model’s assumptions. It earns its place in Chapter 9.

5.3 Wire format

$$\text{Frame} := \{v, t, \text{src}, \text{base_seq}, n, \text{recs}[], \text{auth}\}$$

in canonical CBOR, with the authentication block shaped by placement. The signature input is the canonical bytes of the covered region, and test vectors are frozen. The lean format keeps that signature input and changes only how the records travel; its field table is Table 3.2.

► *TO ADD: a byte-level worked example of one lean frame, hex beside fields.*

5.4 The two-layer data pipeline

The pipeline splits deliberately. **Measured** artifacts — hardware timings, simulation outputs — are non-deterministic, so they are measured once under controlled conditions, committed with environment and configuration-hash headers, and never silently re-rolled.

Derived artifacts are byte-stable functions of those, so an automated gate re-computes every one of them on every run and fails if a single row differs.

This buys reproducibility at the price of *silent staleness* risk: a frozen artifact can go stale when an upstream decision changes and the artifact is not re-run. That is not hypothetical — it is exactly what produced the project’s first recorded defect. The gate exists because the risk is real.

Chapter 6

Experimental Methodology

► *STATUS: DRAFTED from docs/04. The statistical rules below are binding on every result in Chapter 7–Chapter 10.*

6.1 Binding statistical rules

Every experiment obeys the following, and where a result predates a rule that is recorded rather than retrofitted:

- At least 30 seeded repetitions per configuration, with seeds logged in the artifact.
- Report median and bootstrap 95% confidence interval over 10^4 resamples.
- Microbenchmarks: $\geq 10^4$ iterations after 10^3 warm-up, garbage collection disabled, runs spanning a suspend discarded.
- Comparisons state effect size; never bare means.
- Raw artifacts are immutable once frozen.
- Any model-versus-measurement deviation above 10% must be investigated and either explained in writing or the model corrected. **Silent tolerance widening is forbidden.**

6.1.1 Why 30 seeds, and what it does not buy

The 30-seed rule is not arbitrary and is not original policy — it was imposed *after* four headline numbers were found to have been distorted by small-sample means evaluated against thresholds. None was a modelling error; every one was sampling.

Remark (The limit of the rule). More seeds fixes exactly one class of defect: a small-sample mean landing on the wrong side of a threshold. It does nothing for a threshold applied to

a mean rather than a distribution, for an unverified constant on the measurement path, for a configuration change that perturbs the random realisation, or for a claim wider than the experiment supports. Chapter 11 catalogues instances of each, several found in this project’s own work *after* the 30-seed rule was in force.

6.1.2 Thresholds must be applied to distributions

A criterion of the form $V \geq 0.95$ is a *reliability* requirement. Applying it to a mean across seeds answers a throughput question instead. Where this thesis reports a threshold crossing it reports the distribution alongside — minimum, maximum, standard deviation, and the count of seeds that individually fail — and where the mean and per-realisation readings disagree, both are given.

6.2 Pre-registration

Where a run tests a prediction, the prediction is committed to version control *before the data exist*, as its own commit, so the ordering is third-party checkable in the repository history.

This practice is enforced rather than aspirational, and it has cost the project a claim: one pre-registration assertion was *withdrawn* rather than reconstructed, because writing an expectations file after results are known manufactures evidence. It has also exposed a subtler defect in this author’s own practice, recorded in Chapter 9: a pre-registered *directional* prediction that the experiment had no statistical power to resolve.

► *TO ADD: a table of every pre-registration in the project with its commit hash, the prediction, and the outcome. The material exists in docs/; the table does not.*

6.3 Experiment matrix

Exp	Question	Primary output
E1	overhead dominance (T1)	bytes/record, φ per encoding
E2	batching cure (T2/T2a)	φ and bytes/rec vs b ; binding-ceiling check
E3	loss frontier (T3)	measured V vs $(1 - p)^n$; goodput; Pareto
E4	scheme selection (T4)	energy/record, verification wall, crossover
E5	co-design end-to-end (T5)	bytes/energy/goodput vs baselines
E6–E7	low-rate arm and its co-design	feasibility partition, Pareto front
E8	capacity envelope from one load ceiling (superseded)	channel utilisation
<i>Added October 2026, on the frame built as one object</i>		
E9	what a frame is made of	field-level byte budget, both formats
E10	loss with the decoder in the loop (T3')	measured V against keyframe interval, th
E11	the design ladder	bytes, $N_{\max}$, receiver CPU, energy per ru
E12	which link carries a signed, chained frame (T6)	exclusion matrix, twelve data rates
E13	freshness with every term	fill, channel, verification against $D_{\max}$
E14	the low-rate batch at its own record spacing	records per frame, sustainable rate
E15	other PHY rates (model only)	saturation bound against rate
E16	sender energy, metered against modelled	μJ per record
E17	classical stream-signing schemes (model only)	authenticator bytes, frames/s, saturation
—	direct search for $N_{\max}$ (simulation)	delivery against N , bootstrap interval
—	record sizes on public flight logs	keyframe and delta sizes against the gene

Baselines in E1–E8 are A+JSON (naive), A+CBOR (compression-only) and D over-aggregated. From E11 on the comparison is a ladder in which each rung changes one thing, because A+CBOR signs every record *and* sends one record per frame and so confounds two effects.

6.4 Simulation campaign

Two scenarios are used, for two different questions. The *saturation* scenario validates the channel models: IEEE 802.11a ad-hoc at 6 Mbit s^{-1} , RTS/CTS off, with N saturated senders in *both* unicast and broadcast modes, frame sizes taken from the real framer output. The *delivery* scenario measures capacity: the same channel with each node offering one frame per period, so the network runs at a chosen load and the delivered fraction is read at the receivers. The capacities of Chapter 8 come from the second, with the send instant redrawn in every period for the reasons given in Section 8.5. Transport is a packet socket rather than a flow monitor, because MAC-level goodput with no addressing artifacts is what the models predict.

Remark (The comparison rule). Each mode is compared *only* to its matching analytic variant: unicast to the acknowledged DCF model, broadcast to the published broadcast

model. Mixing them is the single most effective way to fabricate a model gap, and this project did exactly that for several phases before catching it.

6.5 Hardware campaign

Two Raspberry Pi 4 units, headless with the performance governor pinned. Timings are re-run from the microbenchmark suite; energy uses external current sensors on a separate measurement host with a GPIO synchronisation line, so the benchmarked device is not instrumented by itself.

The energy harness prints its *predicted* value before reading the meter, deliberately, so a measurement cannot be quietly fitted to expectation.

► *TO ADD: the measurement uncertainty budget for the energy rig. The residual model gap is stated in Chapter 9 but its error bars are not derived.*

Chapter 7

Results I — Bytes, Placement and Loss

► *STATUS: DRAFTED. Every number here is reproduced from a committed artifact by an automated gate; figures are regenerated from frozen data.*

7.1 E1 — authentication dominates after compression

Thirty seeds of one thousand seeded telemetry records per encoding, inline placement, $g_a = 64$ B.

Table 7.1: Measured record size and authentication fraction. Compression moves the bottleneck.

encoding	mean bytes	φ at $g_a=64$	role
JSON	191.09	25.1%	naive baseline
CBOR	66.25	49.1%	compression-only baseline
MessagePack	65.16	49.6%	—
delta-CBOR	45.00	58.7%	the co-design optimum

This is Theorem 4.1 instantiated: at delta encoding the signature is *larger* than the record it signs. The motivating claim of the thesis is a measurement, not a conjecture.

Remark (What s contains). Every size in Table 7.1 includes the record’s own chain link: 32 B in the binary encodings, 64 hexadecimal characters in JSON. The delta record is 13 B of telemetry and 32 B of link, so the link is 71% of what the table calls the record. Counted with the signature, authentication material is 96 of 109 B, 88%, not 58.7%: φ as defined understates the point it was introduced to make. An external review made this observation, and it is what the lean format acts on (Section 7.5): the link is carried once per frame.

Remark (s is not a property of the encoding alone). The sizes above are measured over

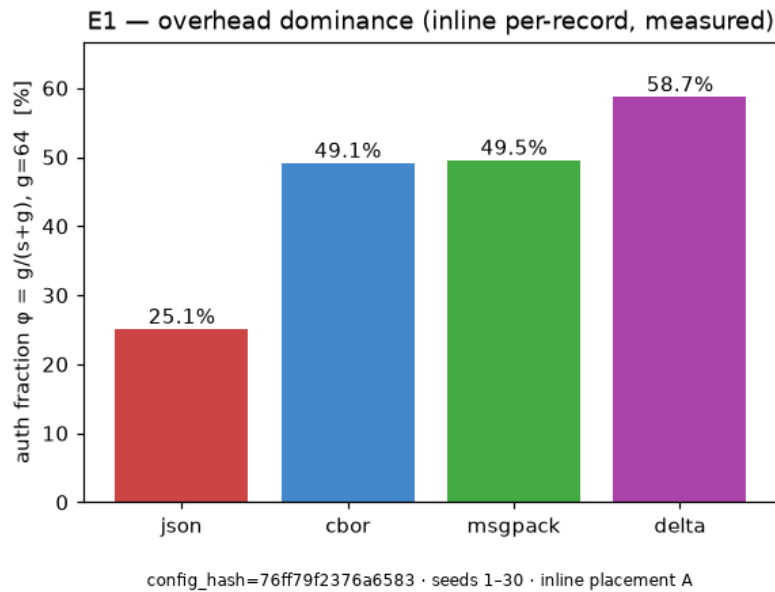


Figure 7.1: Authentication fraction across encodings. Every byte saved on the payload raises the fraction spent on cryptography.

the first ~ 50 seconds of each simulated flight. Sequence numbers and timestamps grow with the record index, and variable-length encodings charge for the extra digits, so JSON, CBOR and MessagePack all grow by 2–4B over a ten-times-longer window while delta-CBOR stays flat — it encodes differences. The bootstrap interval quoted for CBOR is ± 0.02 B and describes *seed* variation; the systematic dependence on window length is ≈ 2.7 B, roughly $150\times$ wider. The direction is conservative: a longer flight inflates the baseline and leaves the optimum alone, so the reported saving is the pessimistic end.

7.2 E2 — batching is the cure, and which ceiling binds

Sweeping placement A against B across encodings and MTU budgets confirms Proposition 4.2 qualitatively and Theorem 4.3 quantitatively. At $M = 256$ the MTU binds and amplification is operative; at $M = 576$ and $M = 1500$ freshness binds and the marginal value of compression is 1.0000 to twelve decimal places.

The practical consequence. On 802.11 the amplification law that motivates aggressive compression is never operative. Compression still pays — it pays exactly $1\times$.

7.3 E3 — the loss frontier

Measured verifiability against batch size and loss probability, placement B versus D. Frame-level batching Pareto-dominates block-level for all $V > (1 - p)^2$, and at $\varepsilon = p$ block-level is infeasible beyond one frame, as Theorem 4.4 requires.

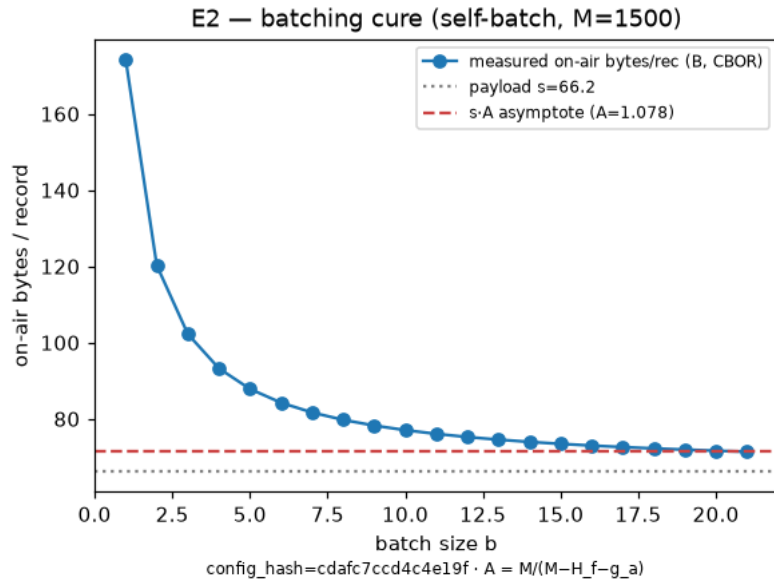


Figure 7.2: Per-record cost against batch size. The curve flattens where the freshness ceiling takes over from the frame-size ceiling.

Remark (What this experiment does and does not validate). The emulator implements independent per-frame loss, which is also what the closed form assumes. Agreement is therefore a *consistency check*, not a validation of the loss model. What rescues the ordering is Theorem 4.5, which holds for any stationary loss process, plus Gilbert–Elliott simulation at matched mean rate showing V_D approaching but never reaching $1 - p$.

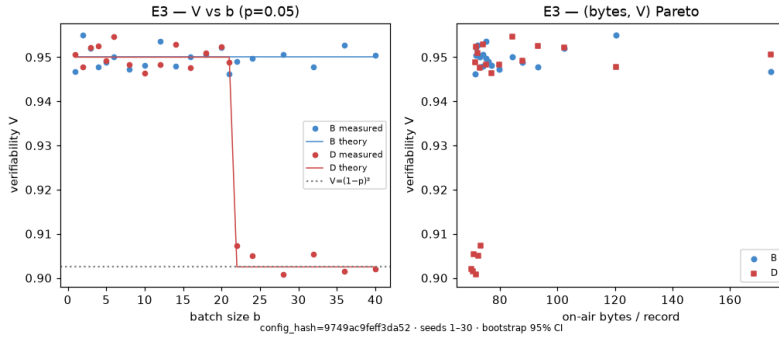


Figure 7.3: The loss-robustness frontier. Block-level authentication trades verifiability for bytes at a rate the constraint does not permit.

7.4 E10 — loss with the decoder in the loop

E3 compares one-frame placements with a block that spans frames. It does not exercise the case Theorem 4.6 describes, because in E3 every frame of placement B is assumed to verify alone. With delta coding that is true only of a frame that starts with a keyframe.

E10 removes the assumption by removing the model: records go through the lean sender, a loss process, and the lean receiver, and V is the fraction of generated records that the receiver decoded, verified and stored. Thirty seeds of 7 200 frames per cell in all.

Table 7.2: Verifiability at frame loss $p = 0.05$ with one self-contained frame in r ($b = 4$). Mean and 95% bootstrap interval over thirty seeds, against Theorem 4.6.

r	B/record	V measured	theorem	$V \geq 0.95$
1	43.25	0.952 [0.947, 0.957]	0.950	yes
2	41.38	0.921 [0.916, 0.927]	0.926	no
4	40.44	0.876 [0.863, 0.889]	0.881	no
16	39.73	0.655 [0.618, 0.690]	0.665	no

The design as first published is the $r = 4$ row: 0.876, not 0.95. The theorem predicts the measurement in every row, and at every other loss rate tried (p from 2.3×10^{-4} to 0.10). With bursty loss of the same mean — a Gilbert process with bursts of four frames — the $r = 4$ design verifies 0.929: better, and still short.

Remark (What this experiment validates, and what it does not). Unlike E3, agreement here is not a tautology: the closed form counts frames, while the experiment runs a codec and a receiver that know nothing of the formula, and they agree. What it does not validate is the loss process itself, which is synthetic at the frame level in both.

The corrected design uses $r = 1$. It costs 3.5 B per record more than $r = 16$ would, and it is the only interval that meets the target at the design’s own loss rate.

7.4.1 When loss grows with frame length

Everything above gives every frame the same loss probability. At a fixed bit error rate a frame of L bytes with o bytes of MAC overhead survives with probability $(1 - \beta)^{8(L+o)}$, so two frames on one channel are related by

$$p(L) = 1 - (1 - p_{\text{ref}})^{(L+o)/(L_{\text{ref}}+o)},$$

and a batch, being a longer frame, is lost more often than the one-record frame it replaces. E10 has a third loss process for this: the bit error rate is set so that the *one-record* frame of the lean format is lost with probability p , and each frame is then lost by its own length.

On a channel that loses 5% of one-record frames, the design’s frame is lost with probability 0.057, and 0.943 of its records are expected to verify (measured: 0.947 [0.941, 0.952]). **That is below the target the design meets at equal frame loss.** It reaches 0.95 only while one-record frames are lost at most 0.044 of the time. The loss budget $p = \varepsilon$ had no margin (Chapter 4); under this reading the batch spends margin it does not have. The byte and capacity comparisons of Chapter 8 are at equal frame loss and carry this qualification.

7.5 E9 — what a frame is made of

Table 3.2 gives the byte budget of both formats. Three things in it matter for everything downstream, and none was visible while sizes were quoted as single numbers.

A delta is not a constant of the encoder. It encodes a difference, and a difference grows with the time it spans:

spacing between records	50 ms	0.1 s	1 s	5.5 s	16.5 s
lean delta record (B)	9.0	10.1	12.5	13.6	16.2

At the 50 Hz of the 802.11 arm the smallest value applies. On the low-rate arm records are seconds apart, and using the 50 ms figure there overstated what a frame could carry (Section 10.3).

Where the chain link sits decides the floor. Figure 7.4 draws bytes per record against batch size for three layouts. With a link in every record, the first format cannot fall below its delta record however large the batch: at 12 records per frame it still costs 54.3 B each. Moving that link to the frame saves $32(1 - 1/b)$ B per record, 24 B at $b = 4$, and gives 51.0 B — less than a per-record link reaches at any batch drawn. The lean format’s shorter header and record coding account for the remaining 7.7 B, to 43.25 B. The two first-format curves are sums of measured parts; the lean curve is emitted frames.

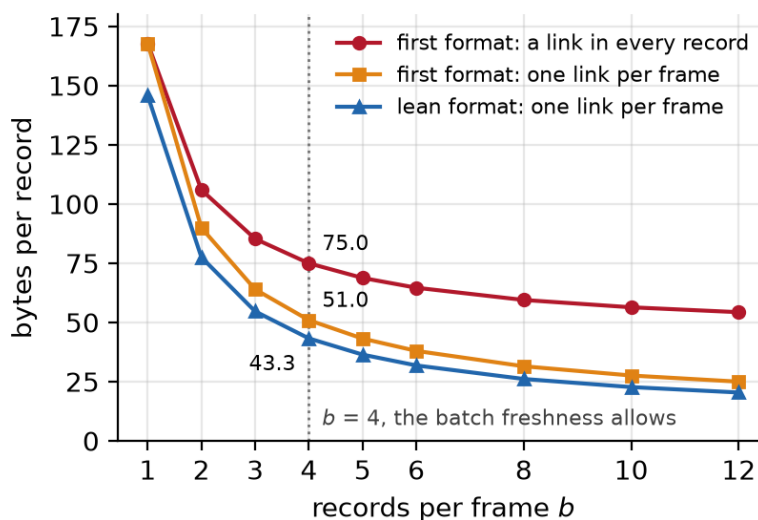


Figure 7.4: Bytes per record against batch size, by where the chain link sits. Records are delta-coded in all three and a frame starts with a keyframe, so at $b = 1$ the first-format curves start below row 1 of Table 8.1, whose records are plain CBOR. The dotted line is the batch that freshness allows at the adopted operating point.

The generator’s delta is the floor of the format. Every one of 29 970 delta records at 50 ms spacing is exactly 9.0 B: nine fields, one byte each. A reviewer called that out as looking synthetic, and it is. To find out whether it matters, the same encoder was run over twelve public PX4 flight logs — four quadrotors, two hexa- or octorotors, three fixed wings and three VTOLs, 29 092 in-flight records — chosen by a rule committed before any log was opened.

Table 7.3: Record sizes on real flight logs against the generator, at matched spacing. The logs carry position at 5 Hz, so nothing finer than 0.2 s can be measured from them.

	keyframe	delta, 0.2 s apart	B/record at $b=4$	saving
twelve flight logs (mean)	22.0	11.1	44.3	69.1%
generator	24.0	10.8	44.6	69.4%

At matched spacing the generator is within a quarter of a byte of real telemetry. The spread is by speed: the six multicopters, mostly hovering, average 10.0–10.5 B per delta, and the three fixed wings up to 13.2 B, for which the saving is 68.2%. So the mean flatters a fast swarm by about a percentage point, and the 50 Hz spacing of the adopted operating point is not covered by any public log found.

Remark (What was predicted, and what was not testable). The prediction committed in advance was a mean delta of at most 13 B and a saving of at least 68%, *at 0.1 s spacing*. The logs turned out to carry position at 5 Hz, so the prediction was applied, unchanged in its numbers, at 0.2 s — the stricter reading, since a difference over a longer interval is never smaller — and holds there. The 0.1 s prediction itself is recorded as untested. One predicted range was missed: the keyframe, predicted at 22–25 B, measures 22.0 B, because multicopters fly slowly enough for one-byte velocities.

Chapter 8

Results II — Co-design and Capacity

► *STATUS: DRAFTED; rewritten October 2026. The capacity results are now direct simulations with intervals. The envelope this chapter used to present, computed from one load ceiling, is kept in Section 8.6 as the approximation it turned out to be.*

8.1 E5 — the end-to-end configuration, and a correction to it

Exhaustive search over encoding, scheme, placement and batch returns delta coding, Ed25519, one signature per frame, $b = 4$. As first reported, that configuration cost 72.0 B per record against 174.3 B for the compression-only baseline: a 58.7% reduction.

Remark (That figure belonged to a design that did not meet its own constraint). The 72.0 B frame amortised one keyframe over four frames. By Theorem 4.6 such a design verifies 0.881 of its records at $p = 0.05$, not 0.95; measured with the decoder in the loop, 0.876 (Section 7.4). The search had admitted it because the verifiability model it consulted assumed every frame decodes alone. With that assumption replaced by the theorem, the search returns the same placement and batch with a keyframe in every frame.

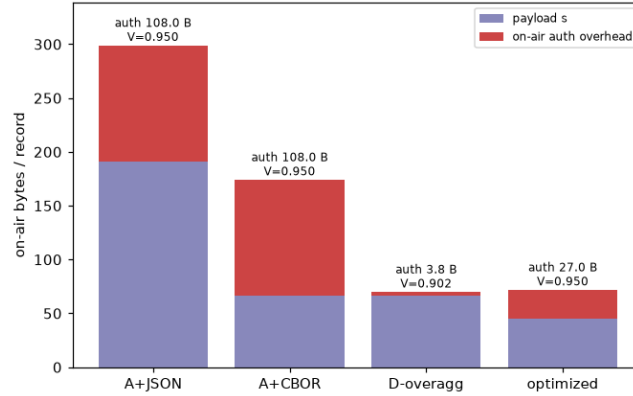
Table 8.1 takes the design apart. A baseline that signs every record and sends it alone confounds two things — sharing a frame, and sharing a signature — so rows 2 and 6 separate them. The corrected figures are 74.96 B per record in the first format, 57.0% below its baseline, and 43.25 B in the lean format, 70.3% below its own.

Remark (Report the decomposition, never the bare authentication ratio). The cut in authentication bytes alone is 75.0%, and it is tempting to lead with it. It must not be led with, because it reduces algebraically to $1 - 1/b$: invariant to the header, the signature size, the encoding and the scheme. It is the definition of batching, not a result.

Table 8.1: The design ladder at 50 records/s and $D_{\max} = 100$ ms. N_{sat} : offered load within the saturation throughput of the broadcast model. $N_{\max}$: largest neighbourhood with mean delivery ≥ 0.95 in simulation, with its 95% bootstrap interval. CPU: share of one receiver core spent verifying at $N_{\max}$. [†] size is a sum of measured parts; rows 5, 6 and 8 are emitted frames.

configuration	b	B/record	with cert.	N_{sat}	$N_{\max}$	CPU
<i>First format: text keys, a chain link in every record[†]</i>						
1 sign every record, one per frame	1	174.25	213.05	18	32 [32, 32]	41%
2 sign every record, share the frame	4	141.25	150.95	24	56 [56, 56]	72%
3 one signature per frame	4	93.25	102.95	30	76 [75, 76]	25%
4 ... and delta records	4	74.96	84.66	34	88 [88, 89]	29%
<i>Lean format: integer keys, one chain link per frame</i>						
5 sign every record, one per frame	1	145.77	184.57	20	35 [35, 35]	45%
6 sign every record, share the frame	4	93.00	102.70	30	76 [76, 77]	98%
7 one signature per frame, no delta [†]	4	54.50	64.20	42	108 [107, 109]	36%
8 one signature per frame, delta (design)	4	43.25	52.95	52	124 [124, 125]	42%

E5 co-design (T5): optimized cuts auth bytes 75.0% vs A+CBOR at $V \geq 0.95$, $p=0.05$ [PASS]



optimized = delta+ed25519+placement B b=4 · timings and both power figures measured on Raspberry Pi 4 (D8)

Figure 8.1: Per-record cost of the configuration as first reported, against the two baselines (first format; see the correction above).

8.2 The factorial ablation, which narrows the claim

A full factorial ablation over the four axes asks whether “co-design” is doing work that single-axis optimisation would not.

- **Placement and batching couple exactly**, by $g_a(1 - 1/b)$ — hence *exactly zero* at $b = 1$, where placements A and B are byte-identical on a single-record frame.
- **Encoding is separable in the first format.** Every interaction term is exactly zero, because s enters additively and therefore *cannot* interact.
- **The scheme axis is byte-degenerate** among the 64 B schemes: they differ only in CPU cost.

An earlier claim that a smaller payload increases the value of batching was a **ratio-scale artifact**: the absolute saving is 81.0 B for every encoding. The four-axis co-design claim was overstated and is now precise.

Remark (One coupling the first-format ablation could not see). In the lean format encoding and batching *do* interact, in one specific way: a record can be coded as a delta only if the record before it is in the same frame or is known to have arrived. Row 6 of Table 8.1 is smaller than four copies of row 5 partly for that reason. The separability result above is a property of the first format, where each record is encoded alone.

8.3 E4 — scheme selection, and the receiver’s CPU

On ARM — the deployment platform — Ed25519 verification is $33.2\times$ cheaper than a BLS pairing, which keeps Ed25519 optimal for self-batching at any plausible radio/CPU power ratio. The ordering between Ed25519 and ECDSA-P256 *flips* between x86 and ARM, so the choice is platform-specific; the byte result is identical either way, since both are 64 B.

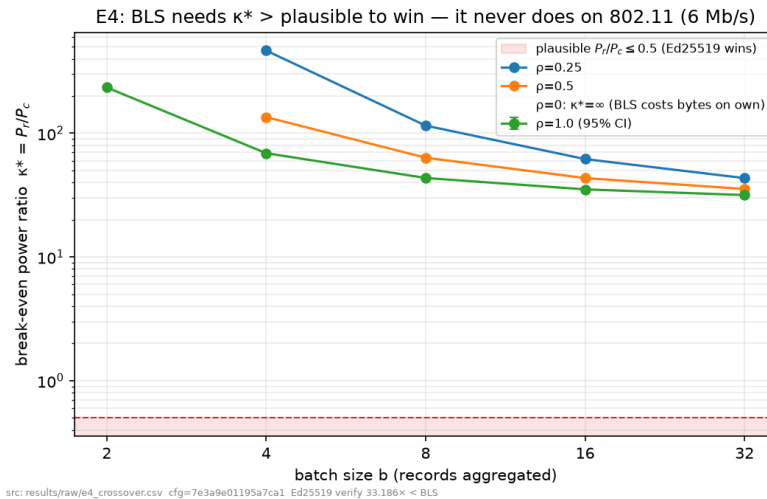


Figure 8.2: The scheme crossover. BLS becomes worthwhile only in the cross-signer regime.

A receiver verifies every neighbour’s frames, and that is a second ceiling on the neighbourhood. With k signatures per frame, verification time t_v and one hash per record t_h , one neighbour costs $(\Lambda_i/b) k t_v + \Lambda_i t_h$ seconds of CPU per second, and one core serves

$$N_{\text{cpu}} = 1 + \left\lfloor \frac{1}{(\Lambda_i/b) k t_v + \Lambda_i t_h} \right\rfloor$$

nodes. On the Raspberry Pi 4 that is 296 neighbours when each frame carries one Ed25519 signature and 77 when every record is signed. At its N_{max} the design uses 42% of a core, so the channel binds first. Row 6 of Table 8.1 shows the constraint is not idle: with a signature on every record it reaches the channel’s limit at 98% of a core. With BLS a

verification takes 8.61 ms and one core serves 10 neighbours: BLS is excluded by the CPU long before the channel is reached, in addition to its larger signature.

These figures verify one signature at a time. Ed25519 was designed for batch verification: its authors report under 134 000 cycles per signature in a batch of 64, against 273 364 for one [5], which would roughly halve the receiver’s cost. An external review named it among the stronger alternatives this work had not considered. It is not used here — the cryptographic library this implementation is built on exposes no batch interface — so N_{cpu} is conservative for Ed25519 by about that factor, and measuring it on the board is open. ECDSA does not admit it.

8.4 E17 — the classical stream-signing schemes, placed

Chapter 2 describes the schemes that amortise one signature over many packets. None is implemented in this thesis. Table 8.2 places each at the adopted operating point as *the lean one-record frame with that scheme’s authenticator in place of the signature* — the same header, chain link and record — with the authenticator sized from the scheme’s own definition and this thesis’s primitives (32 B digests, 64 B signatures). Two capacities are given: the saturation bound N_{sat} , and N_{max} from the direct search of Section 8.5 on a frame of that size at that rate — five more cells, whose crossings were predicted and committed before they were run (Section 8.5.1).

Table 8.2: Stream-signing schemes at 50 records/s. Sizes are modelled; N_{max} is simulated for a frame of that size, with its 95% bootstrap interval. [†] no non-repudiation: a third party cannot be shown who sent a record; [‡] a packet does not verify alone, so N_{max} — frames delivered — is an upper bound on records verified.

scheme	auth. B/record	frames/s	total B/record	N_{sat}	N_{max}
MAVLink 2 tag [†]	13	50.0	93.8	23	42 [41, 42]
TESLA ^{†‡}	24	50.0	105.8	22	40 [40, 40]
Gennaro–Rohatgi [‡]	32	50.0	113.8	21	39 [39, 39]
EMSS [‡]	65	50.5	147.3	20	34 [34, 34]
Wong–Lam tree, block of 4	129	50.0	210.8	17	28 [28, 29]
a signature on every record	64	50.0	145.8	20	35 [35, 35]
this design, $b = 4$	16	12.5	43.3	52	124 [124, 125]

Every one of these schemes spreads an authenticator across packets and still sends a frame per record. A frame costs 173 μs of channel time before its first payload byte, so the weight of the authenticator hardly matters: replacing a 64 B signature by a 13 B tag — and giving up non-repudiation — moves N_{sat} from 20 to 23 and the simulated N_{max} from 35 to 42. Four records in a frame move them to 52 and 124. The stream-signing literature amortises the *cost of signing*; the cost that binds on a contended channel is the *frame*.

Three things the table does not show. Tree chaining is the one classical scheme that

keeps both properties a ledger needs — every packet verifies alone and its origin can be shown to a third party — and it pays 129B per packet for them. Batching has a cost no per-packet scheme has: a lost frame takes b records with it, and records wait for the batch to fill. And a scheme with a shorter frame is lost less often at a given bit error rate (Section 7.4), which this table does not credit.

8.5 Capacity, by direct search

Bytes matter because they are airtime, and airtime decides how many neighbours can be heard. The $N_{\max}$ column of Table 8.1 is the answer to the question an operator asks — *how many UAVs, at what telemetry rate?* — and it is found by simulating each configuration at its own frame size and frame rate over a grid of neighbourhood sizes, thirty seeds per point, and reading off where mean delivery leaves 0.95. The interval is the 2.5–97.5 percentile of that crossing over ten thousand bootstrap resamples of the seeds.

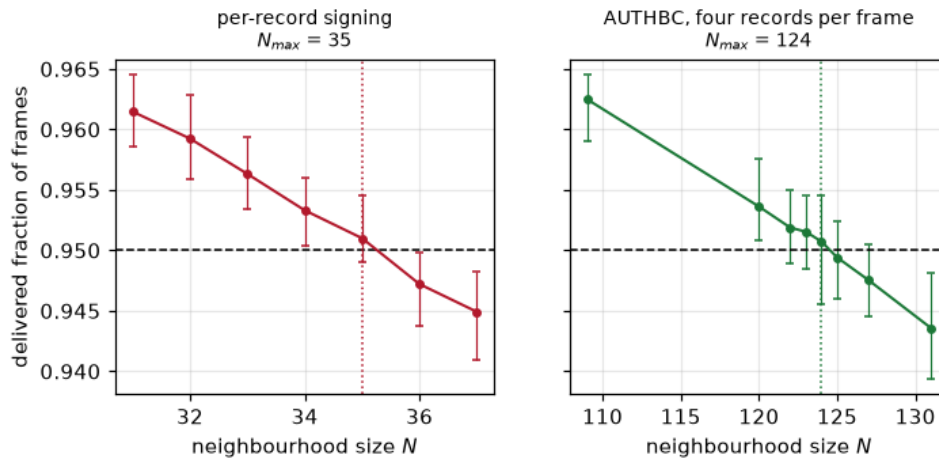


Figure 8.3: Delivered fraction of frames against neighbourhood size, lean format: per-record signing and the design. Mean of thirty runs; bars span the runs.

Table 8.3: $N_{\max}$ of baseline and design at both operating points, with 95% bootstrap intervals.

	per-record signing	design	ratio
<i>50 records/s, 100 ms (compliant with TS 22.125)</i>			
first format	32 [32, 32]	88 [88, 89]	2.8×
lean format	35 [35, 35]	124 [124, 125]	3.5×
<i>20 records/s, 250 ms (relaxed)</i>			
first format	78 [77, 78]	219 [215, 221]	2.8×
lean format	85 [85, 86]	306 [306, 308]	3.6×

These figures hold the *mean* over runs to the 0.95 target, which is the headline criterion throughout this thesis. The stricter reading — that 95% of individual runs reach it —

gives 34 and 120 in the lean format and 31 and 86 in the first: at most four nodes fewer, now that a run samples the channel and not one phase configuration. On the low-rate arm the two readings are far apart (Chapter 10).

Remark (Do not round this to one number). The advantage of the design is not one number. It hardly moves with the operating point — $3.5\times$ and $3.6\times$ in the lean format, $2.8\times$ and $2.8\times$ in the first — and it depends on the format, because the first format’s records each carry a chain link that no batch amortises. Quote it with its format. An earlier draft said “ $\approx 3\times$ holds across readings”; the four ratios it rested on came from the single load ceiling, ran from 1.9 to 3.2, and are superseded with it.

The larger step is sending a quarter as many frames. Row 6 of Table 8.1, which shares the frame and keeps a signature per record, already reaches 76 neighbours ($2.2\times$ the baseline); sharing the signature as well adds $1.6\times$. Every frame pays a preamble, an inter-frame space, a backoff and MAC framing whatever its length — about $173\mu\text{s}$, 47% of what a baseline frame costs the channel — so removing three frames in four is worth more than their bytes.

8.5.1 A rule for the crossing, and the test it was put to

The envelope this chapter used to report applied one utilisation ceiling to every configuration (Section 8.6), and Chapter 9 records how that failed. Looking at the six crossings it failed on suggested something simpler: at the 5% level, loss is a line in airtime. With $f = \Lambda_i/b$ the frames per second a node sends and T the time one frame holds the medium (its airtime plus DIFS),

$$0.05 = Nf(aT + c) \quad \implies \quad N_{\max} \approx \frac{0.05}{f(aT + c)}. \quad (8.1)$$

NfT is the fraction of time the neighbourhood’s frames occupy the medium, so the first term says loss grows in proportion to occupancy; the second, with $c = 8\mu\text{s}$ fixed at twice the simulator’s preamble-detection time, is the chance that another frame starts inside the window in which a station cannot yet hear it. Only a is fitted.

A line found by looking at six points is a hypothesis. It was treated as one: the seven configurations that had not been simulated were named, the line’s value for each was written down with a tolerance of $\pm 6\%$, and both were committed before any of the seven was run. The slope is $a = 0.071$, the mean of the six per-configuration values, which reproduces those six within 3.8%.

The prediction held. All seven crossings lie within 3.4% of the line, and in each of the five configurations where the two rules disagree by more than a tenth the simulation is closer to the line than to the ceiling, which misses the thirteen simulated configurations by up to 15%.

Table 8.4: The seven held-out configurations: the crossing the line predicted, the crossing simulated (interpolated, thirty seeds per node count), and the deviation of the simulation from the line and from the single load ceiling.

configuration	frame, rate	line	simulated	vs line	vs ceiling
first, one signature per frame	373 B, 12.5/s	78.4	76.2	−2.7%	−3.5%
lean, one signature, no delta	218 B, 12.5/s	110.3	108.9	−1.3%	−11.5%
first, design as first specified	288 B, 12.5/s	93.5	91.3	−2.3%	−8.7%
relaxed: first, per-record signing	174 B, 20/s	78.1	78.4	+0.3%	−11.0%
relaxed: first, design	300 B, 5/s	227.6	219.8	−3.4%	+5.7%
relaxed: lean, per-record signing	146 B, 20/s	84.9	85.5	+0.7%	−15.3%
relaxed: lean, design	173 B, 5/s	312.4	308.1	−1.4%	+14.5%

A second test, on frames of another kind. The stream-signing baselines of Table 8.2 are five more frames, all shorter or longer than any the line had seen and all at one frame per record. The same procedure was followed: the line’s crossing for each, with the slope unchanged, was committed before the runs.

scheme	frame, rate	line	simulated	vs line
MAVLink 2 tag	94 B, 50/s	40.6	42.2	+3.8%
TESLA	106 B, 50/s	38.8	40.4	+4.2%
Gennaro–Rohatgi	114 B, 50/s	38.0	39.4	+3.8%
EMSS	146 B, 50.5/s	33.6	34.8	+3.5%
Wong–Lam tree	211 B, 50/s	28.0	29.0	+3.5%

All five are inside the $\pm 6\%$ band, and all five are *above* the line, by 3.5–4.2%. That offset was written down beforehand as an expectation, not a prediction: the two calibration configurations at 50 frames per second sat 3.4 and 3.8% above the line, so if the residual belongs to the frame rate these five would too. They do. The line’s error is therefore not noise at this rate; it is a small, repeatable bias, and it is the reason the remark below forbids quoting the line as a capacity. These cells do not separate the line from the ceiling — at this frame rate the two differ by a few percent — which the registration said in advance.

Remark (What this licenses, and what it does not). Equation (8.1) is an empirical fit, not a model. It holds at one loss level — loss is convex in load, so the line says nothing about a 1% or a 10% target — at one PHY rate, in one collision domain, for frames of 94–565 B, 5–50.5 frames per second and 28–306 nodes, and in a simulator. Its residual is ordered by frame rate, from +3.6% at 50 frames per second to −2.4% at 5. Every capacity quoted in this thesis is a simulated value; the line is what one would use to interpolate between them, and to see why a ceiling on *utilisation* could not work: it divides the offered load by the saturation throughput of N stations, and loss in a lightly loaded network has no reason to follow that yardstick.

8.5.2 How the traffic source was chosen

Each simulated node sends one frame per period at an instant drawn uniformly within the period. That choice was not the first one, and how it was reached is part of the result.

The scenario this project had used throughout sends frames *strictly* periodically: each node gets one random start offset and keeps it. In such a run the relative phase of any two nodes is frozen. Per-node output shows what that does: in the worst of thirty runs at $N = 29$, four nodes delivered *none* of their frames while 22 lost none at all. The simulator raises carrier sense only at the end of a 4 μ s preamble-detection period, so two nodes whose phases fall within that window collide in every period of the run; and nodes whose phases put them in the same busy period contend with each other, and only each other, for all of it. A strictly periodic run is one draw from a lottery of phase configurations, and the spread across runs describes the lottery.

Three tests, each with its prediction committed before its runs, established what follows from that. Jittering each frame’s send time, or letting every pair of nodes drift through all relative phases, changes the thirty-run *mean* by less than the 0.005 set in advance — none of sixteen comparisons exceeds 2.5 standard errors — while the spread between runs falls by a factor of ten to twenty when phases are redrawn every period. The runs reported here redraw the phase every period. Two of the predictions made along the way failed, and both failures are recorded with the data: a jitter of 1 ms was predicted to halve the spread and did so at two test points of four, and a phase sweep was predicted to bring it under 0.004 and reached 0.003–0.006.

Remark (“Within tolerance” is not “zero”). When all six cells had been run under both sources, the mean delivered fraction was higher with redrawn phases by 0.003 on average over 14 points, and every one of the six crossings was higher, by 1–9%. The points are close to independent — the simulator hands out random streams in creation order, so runs at different node counts draw different start offsets, and their per-seed results are uncorrelated — which puts the average at about two standard errors: weak evidence of a real difference of that size, and no more. If phases are uniform and a period’s losses depend only on that period’s phases, the expected difference is zero, so a real one would need a mechanism not yet identified. No reported capacity depends on the answer, since all of them use the redrawn source; what depends on it is how the strictly periodic figures of earlier work should be read.

8.6 What the earlier envelope was

Until October 2026 this chapter presented a “capacity envelope” computed, not simulated: the $V \geq 0.95$ crossing had been measured once, at $N = 50$ with a 288 B frame, as a utilisation $U = 2.435$ (offered load over saturation throughput), and that one ceiling was applied to every configuration.

Table 8.5: The envelope as previously reported, from the single load ceiling. **Superseded** by Tables 8.1 and 8.3; kept because the comparison is the finding.

configuration	Λ_i	$D_{\max}$	b	$N_{\max} (U < 1)$	$N_{\max} (V \text{ mean})$	$(V \text{ per run})$
optimized delta/B	50	100 ms	4	35	100	64–91
A+CBOR (compression-only)	50	100 ms	1	18	31	24–29
optimized delta/B	20	250 ms	4	103	213	163–201
A+CBOR	20	250 ms	1	32	88	54–80

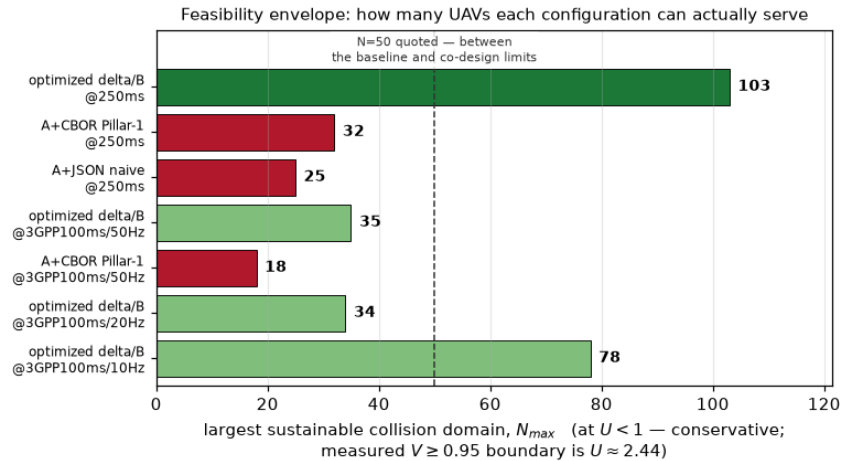


Figure 8.4: The envelope as previously reported (saturation threshold). Superseded for the $V \geq 0.95$ threshold by direct simulation.

An earlier experiment had tested half of the assumption behind that ceiling — that it does not depend on frame size — at $N = 50$ (Chapter 9). The other half, that it does not depend on N , was predicted to hold within $\pm 10\%$ in six configurations, and the prediction was committed before the runs. **It failed.** The lean design crosses near 118–124 neighbours where the ceiling says 142; the first format’s shared-frame configuration crosses *above* the ceiling’s value. The signs differ between configurations, so the ceiling cannot be re-tuned. The reason is not subtle in hindsight: U divides the offered load by the saturation throughput of N stations, which falls as N grows, while the loss of a lightly loaded network depends on how much airtime the traffic occupies and not on how many stations produce it.

The ratios of the old envelope had been called “protected by construction”, because numerator and denominator used the same ceiling. They were protected against an error in the ceiling’s *value*. They were not protected against the ceiling being the wrong *kind* of quantity, and they moved.

Table 8.6: Delay of the oldest record in a frame, $b = 4$, 50 records/s. Channel delay is simulated at the heaviest load that still delivers 95% of frames.

	fill (ms)	channel (ms)	verify (ms)	total (ms)	margin (ms)
Ed25519, mean channel delay	80	0.76	0.26	81.0	19.0
Ed25519, 99th percentile	80	2.40	0.26	82.7	17.3
Ed25519, worst observed	80	3.49	0.26	83.8	16.3
BLS, worst observed	80	3.49	8.61	92.1	7.9

8.7 Freshness, with every term

The freshness model of Chapter 4 was fill time plus airtime. Table 8.6 adds the two terms it left out: channel-access delay, which had been measured and never added, and verification time. Fill time dominates; the oldest record is verified within 83.8 ms of being generated. A fifth record would need a fill time of 100 ms by itself, which is why the batch stops at four.

8.8 Certificates

The rows so far assume that receivers already hold the senders’ public keys. Vehicular practice sends a full certificate in every fifth message and an eight-octet digest in the others [12]. Charging that to every frame — a 162 B certificate, hence 38.8 B per frame on average — gives the “with cert.” column of Table 8.1. Certificates are a per-frame cost, so batching amortises them too, and both the baseline and the design are charged.

These are *explicit* certificates: a verification key and the issuer’s signature over it. IEEE 1609.2 also defines *implicit* certificates (ECQV), which carry only a reconstruction value from which the key is derived with the issuer’s root certificate; the source this accounting follows gives a signed message of at most 226 B with an implicit certificate against 330 B with an explicit one [12]. Implicit certificates are not priced here, so the column is an upper bound on what certificates cost either design. Pricing them is future work; it would lower both rows and leave the comparison between them where it is, since whichever certificate is used is paid once per frame.

Remark (Why the order of this work matters). The certificate term was added to the byte model *before* the published figures it is compared with in Chapter 2 were retrieved. Adding it afterwards would have been fitting the correction to the answer.

8.9 Energy

The batched sender spends 51% less energy per record. The model composed from separately measured times and powers is 7.5–12.3% low, the gap being frame assembly in

Table 8.7: Sender energy per record on a Raspberry Pi 4, first format: metered (median and range of five repetitions) against the composed model.

	metered (μJ)	model (μJ)	gap
every record signed	118.8 [118.5, 119.9]	105.8	12.3%
one signature per four records	58.4 [58.1, 59.3]	54.3	7.5%

an interpreted prototype. Only the sender is metered, and only the first format’s encoders: the lean codec has not been timed on the board, and receiver and radio energy are modelled.

Chapter 9

Model Validation and Hardware

► *STATUS: DRAFTED.*

9.1 Three independent implementations of the same physics

The broadcast channel model is the load-bearing input to the capacity envelope, so it is validated against two independent sources rather than one.

Table 9.1: Saturation throughput (Mb/s), 1400 B broadcast frames. The slot-exact simulator was written *before* the published model was located and shares none of its assumptions.

N	published model	slot-exact sim	Δ	NS-3	Δ
5	4.3329	4.3305	−0.06%	4.3318	−0.03%
10	3.1319	3.1321	+0.00%	3.1288	−0.10%
20	1.7823	1.7830	+0.04%	1.7866	+0.24%
35	1.2407	1.2407	−0.00%	1.2343	−0.51%
50	1.2595	1.2585	−0.08%	1.2541	−0.43%

Agreement is $\leq 0.08\%$ against the independent simulator and $\leq 0.51\%$ against NS-3. Unicast agrees with the acknowledged DCF model to $-0.40/+1.29\%$ across $N = 5\text{--}50$, with fixed-point residuals below 10^{-12} .

9.2 The mechanism, confirmed by removing it

The reason broadcast needs its own model is a specific asymmetry. A station that has just transmitted redraws its backoff and may draw zero, taking the medium immediately after the interframe space, while every station that deferred necessarily holds a counter of at least one. In unicast the acknowledgement timeout blocks colliders from doing this; in broadcast nothing does.

The slot-exact simulator can disable *exactly* that asymmetry and nothing else. Doing

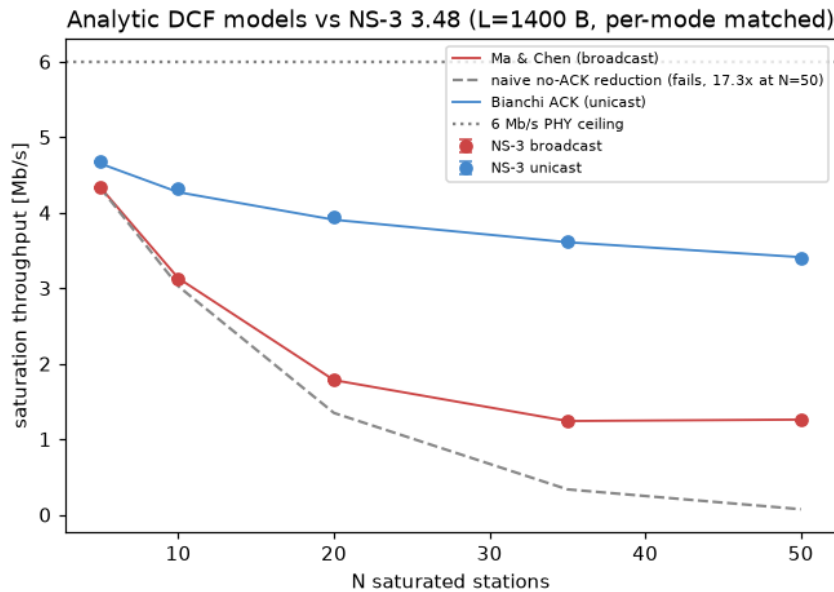


Figure 9.1: Analytic models against simulation, both modes.

so collapses it onto the naive reduction — a $16.9\times$ gap at $N = 50$. **The mechanism attribution is therefore not an inference from agreement; it is a controlled intervention.**

Remark (An expensive lesson). This project used the naive reduction — the unicast model with the acknowledgement removed and the transmission probability fixed — for several phases. It under-predicts by $16\times$ at $N = 50$. The published broadcast model’s own authors open by warning that unicast models cannot simply be reduced for broadcast analysis. *The failure was reading a model without reading its preamble.*

9.3 A non-monotonicity that is real

Saturation throughput is **not monotone decreasing** in N : it dips near $N \approx 35$ and recovers. All three implementations reproduce it, so it is a property of the freeze-process series rather than a bug — each freeze stage contributes a hump peaking at a station count set by the contention window.

This matters operationally. The capacity search breaks on the first violating N , which is valid only if utilisation is monotone. It is — because the explicit factor N in the numerator outruns the recovery — but **that is an arithmetic accident, not a structural guarantee**, and it is now verified exhaustively rather than assumed.

9.4 Hardware

Two Raspberry Pi 4 units in ad-hoc mode at 5 GHz measured broadcast link loss at $p = 2.3 \times 10^{-4}$ (4 frames lost of 17 600) and airtime at 1.995 ms per frame against 1.988 ms predicted, a gap of 0.36% — an independent check on the timing constants underlying every airtime figure in this thesis.

Remark (Two warnings, both earned). This validates *airtime*, not contention: one transmitter means zero contention, so the broadcast contention model remains simulation-only.

And an initial 2.4 GHz sweep produced a tidy 97.45% delivery figure that was **saturation at the 802.11b 1 Mb/s broadcast basic rate**, not channel loss. It was caught by a pre-stated prediction plus a load sweep, and is kept in the record labelled as what it was. Repeatability did not protect against it — the standard deviation across windows was small. *A precisely reproducible measurement of the wrong thing looks exactly like a good measurement.*

9.5 The utilisation ceiling, and a flaw in the author’s own pre-registration

The capacity envelope applies one measured utilisation ceiling to configurations whose frames span 153–299 B, but that ceiling had only ever been measured at 288 B. A second sweep at 174 B — the compression-only baseline frame, which produces the denominators the capacity ratios divide by — tested it, with the prediction committed data-free beforehand.

	288 B	174 B
$V \geq 0.95$ crossing	2.435	2.367
combined standard error	± 0.151	
significance of the shift	0.45σ	

Statistically indistinguishable across a $1.66\times$ change in frame size. At the time this was read as justifying the universal ceiling. It justified half of it, at one N ; the other half is below.

Remark (The prediction that should not have been made). The pre-registration also predicted a *direction* — that the crossing would drift higher — and offered a mechanism. It drifted lower, at 0.45σ . The experiment has no power to resolve the sign, so had the noise fallen the other way the author would have recorded a confirmation he had not earned, and the mechanism offered is neither supported nor refuted.

This is recorded rather than quietly dropped because it is the same defect class the rest of this thesis catalogues: **a number that looks like evidence and is not.** A

directional prediction needs a power estimate, or must be stated as a band. The load-bearing prediction did carry one, which is why the run answers its question at all.

9.6 The other half: invariance in N , tested, and it does not hold

Both sweeps above are at $N = 50$, while the envelope applied the ceiling from $N = 2$ to $N = 213$. For two months that half of the assumption was listed as stated and not measured. An external review asked for confidence intervals on the 802.11 capacities, “as for LoRa”, and the LoRa figure is a direct search in N ; so the 802.11 figure was made one.

The prediction, committed before any run: in each of six configurations, the directly simulated $N_{\max}$ lies within $\pm 10\%$ of the ceiling’s value. No direction was predicted, and a power estimate was given.

The outcome. It fails. With the scenario exactly as published (1080 runs):

configuration	frame, rate	ceiling	direct (interpolated)	deviation
first, sign every record	174 B, 50/s	31	31.0	−0.1%
first, batched design	300 B, 12.5/s	97	87.8	−9.5%
lean, sign every record	146 B, 50/s	34	32.4	−4.7%
lean, batched design	173 B, 12.5/s	142	118.3	−16.7%
first, shared frame	565 B, 12.5/s	49	54.7	+11.7%
lean, shared frame	372 B, 12.5/s	80	74.8	−6.6%

The ceiling overstates the capacity of short frames sent by many nodes and understates that of long frames sent by few. At the crossings U runs from 1.98 to 2.61 where the ceiling says 2.435. It is not a constant, and Chapter 8 no longer uses it.

Remark (Three defects in the measurement itself, each of which could have hidden the result). The driver first reported the *median of the bootstrap replicates* as $N_{\max}$ where the registered estimate is the sample’s own crossing; they differed in one configuration (31 against 29). The grid was coarse — eleven nodes between points in the configuration that decided the outcome — so an interpolated crossing was added afterwards and is labelled as an addition. And the power estimate promised ± 2 nodes where the interval is ± 5 . None changes the verdict; all three are in the record.

What follows from the failure — the traffic-source experiments, and the capacities now reported — is in Section 8.5.

Chapter 10

The Low-Rate Regime and the Exclusion Bound

► *STATUS: DRAFTED, and rewritten in October 2026 after an external review. This chapter carries the thesis’s most durable result and two successive corrections of it. Both corrections are recorded in Section 10.2.5, not folded into the text as if the result had always been stated this way.*

10.1 Why the low-rate arm is a different regime, not a slower one

The 802.11 numbers do not transfer. They differ by two to four orders of magnitude, and the binding constraint differs *in kind*: a regulatory airtime quota instead of a frame size or a latency budget. This is the third regime in the taxonomy of Theorem 4.3:

regime	what caps the batch	where
MTU-limited	frame size	—
freshness-limited	$\Lambda_i D_{\max}$	802.11
duty-cycle-limited	regulatory airtime quota	low-rate

On this arm the arrival rate and the freshness are *derived*, not configured: a frame of airtime T may repeat only every T/duty , so the sustainable rate and the staleness of the oldest record are the same interval. There is no independent deadline to trade against bytes.

This chapter is *analysis and simulation, not measurement*. No LoRa hardware was instrumented and no LoRa energy figure is reported: the only metered radio power in this work is a Wi-Fi figure, and reusing it for a different transceiver would be fabrication. Every LoRa constant is transcribed from the SX1276 datasheet [33] and from LoRaWAN RP002-1.0.3 [19] (Tables 12 and 13; EU863-870 duty cycle below 1%).

10.2 The exclusion

Theorem 4.8 says that a link of maximum payload M admits a frame that verifies alone only if

$$M \geq H_f + L + g_a + s_k,$$

a header, one chain link, an authentication object and one record that decodes without its predecessor. An exclusion is claimed against the *smallest frame the format can emit at all*, not against the frames one telemetry stream happened to produce.

10.2.1 The floor of the format

Take the lean format of Chapter 3 with sender 0, sequence number 0 and one record in which every field is zero. Every integer then occupies its minimum width:

$$\underbrace{16}_{H_f} + \underbrace{35}_L + \underbrace{64}_{g_a} + \underbrace{9}_{s_k} = 124 \text{ bytes.}$$

That frame was emitted, signed and verified; it is not an estimate. **Header, link and signature alone are 115 B.** The first format's floor is higher still, because its record carries its own 32 B link and its keys are text.

10.2.2 All twelve data rates

RP002-1.0.3 defines *twelve* EU863-870 data rates: DR0–DR6 are LoRa, DR7 is FSK and DR8–DR11 are LR-FHSS. (Earlier versions of this chapter considered seven.)

Table 10.1: EU863-870 data rates against one self-contained, signed, chained frame ($g_a = 64$ B). The verdict is the same in both frame formats and under both of the standard's payload tables.

data rates	payload N	verdict	what stops it
DR0, DR1, DR2, DR8, DR10	50 or 51 B	excluded	the signature alone: $64 > 51$
DR3, DR9, DR11	115 B	excluded	header, link and signature fill the payload
DR4, DR5, DR6, DR7	222 or 242 B	feasible	—

Eight of twelve data rates cannot carry one signed, hash-chained frame.

None of the eight depends on the telemetry: each is excluded against the floor. They fall into two groups that are excluded for different reasons, and the difference decides what could ever rescue them.

Five rates: the signature. At 50 or 51 B a 64 B signature does not fit. This holds with a zero-byte header, no chain link and no record. Only a smaller authentication object could change it.

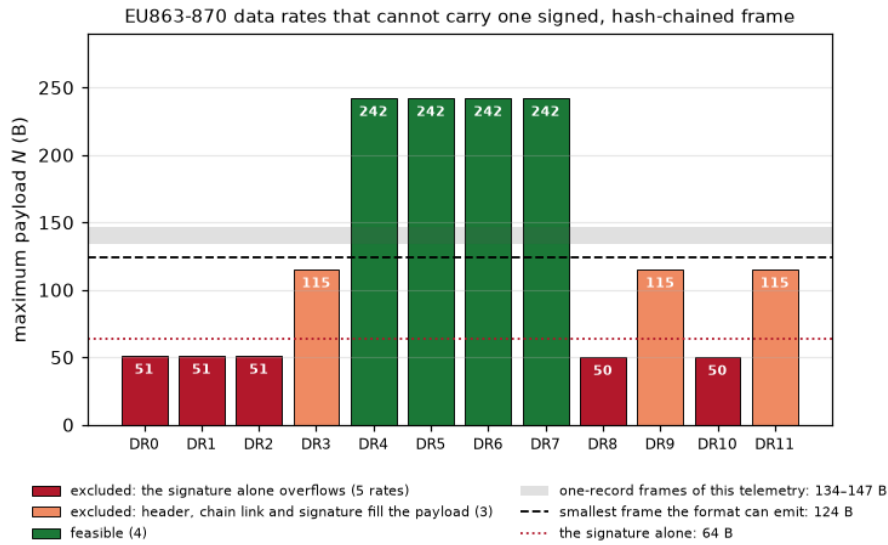


Figure 10.1: Payload limit of each data rate against the floor of the lean format and the frames this telemetry produces.

Three rates: what a frame must contain. At 115 B the signature fits, and so does the header. What does not fit is the chain link together with a record: $16 + 35 + 64 = 115$, the whole payload, before a single byte of telemetry.

Remark (Which payload column). Published EU868 tables differ. Klimiashvili *et al.* list DR3 at 123 B where this work uses 115 B [34]. Both read the standard correctly: 123 is the maximum *MACPayload* and 115 the maximum *application* payload, the two differing by the 8 B of *FHDR* and *FPort*. The application figure is the applicable one, since the frame travels inside *FRMPayload*. Under the other reading the floor of 124 B still does not fit, by one byte — too thin a margin to lean on, which is why it is recorded here and not offered as a second argument.

10.2.3 Scope

The exclusion is a statement about a particular kind of system. It holds under four conditions, and each is necessary.

1. **The limits are LoRaWAN's.** They come from the regional parameters. The LoRa physical layer itself carries 255 B at any spreading factor and is not excluded; a private LoRa network that ignores the regional payload table is outside this result.
2. **Every frame carries a public-key signature.** A symmetric tag is far smaller, and authenticates a link instead of an originator.
3. **There is no loss recovery.** Retransmission or coding across frames changes what it means for a frame to arrive.

4. $\varepsilon \leq p$. This is what forbids spreading a signature over several frames: a unit spanning n frames verifies with probability $(1-p)^n$, so $V \geq 1-\varepsilon$ needs $n \leq \lfloor \ln(1-\varepsilon)/\ln(1-p) \rfloor$, which is 1 exactly when $\varepsilon \leq p$. At $p = 0.01$ and $\varepsilon = 0.05$ a signed unit may span five frames, and the exclusion no longer holds as stated.

10.2.4 What each relaxation buys

Table 10.2: The lean format under four authentication choices. *Floor*: every field zero. *Range*: frames of this telemetry over a flight.

	floor	range	50/51 B	115 B	222/242 B
64 B signature, chain link on air	124	134–147	no	no	yes
... chain link not sent on air	89	99–112	no	yes	yes
48 B BLS signature	108	118–131	no	no*	yes
13 B symmetric tag	72	82–95	no	yes	yes

* a frame of zeros would fit; no frame of this telemetry does.

Three readings of Table 10.2 are worth stating.

The 115 B rates are excluded by the chain link, not by the signature. Remove the link from the air and they fit with ordinary 64 B signatures. That row is arithmetic, not an implemented format: without the link a receiver that has missed a frame cannot rebuild the next one, so removing it trades the exclusion for the dependence that Theorem 4.6 counts.

A three-byte miss is not an exclusion. The smallest standardised signature is a 48 B BLS12-381 point in the minimal-signature-size variant [35], which targets 126-bit security. It fits a 51 B payload and leaves three bytes. At 115 B a frame of zeros would fit (108 B) while the smallest frame of this telemetry is 118 B. That margin is within reach of a format change or a different payload, so the artifact grades it separately and it is *not* counted among the excluded rates.

At 50 or 51 B, hash chaining itself does not fit. Header and link are 51–58 B before any authenticator or record. Even a 13 B symmetric tag, as MAVLink 2 uses, leaves no room. The exclusion there rests on two independent facts.

10.2.5 How this result reached its present form

This section exists because the result was stated wrongly twice, and the second error was made *while correcting the first*.

First statement (July 2026). $s_{\max} = M - H_f - g_a \geq s_{\min}$, with $H_f = 44$ B, $g_a = 64$ B and $s_{\min} = 13$ B, applied to seven data rates. DR0–DR2 were excluded by the signature and DR3 “by six bytes” (7 B of room for a 13 B record): *four of seven*.

First correction (August 2026). The header is 29 B of CBOR text key names; the same seven keys as small integers cost 7 B. An integer-keyed header of 22 B gave DR3 29 B of room, and the headline was changed to *three of seven*, with DR3 “rescued” by a header redesign. The correction was presented, in this chapter and in the paper’s abstract, as an example of composing two facts that had been recorded separately.

Second correction (October 2026). An external review asked what a frame that verifies alone must actually carry. Two things had never been charged in either statement:

- the **chain link**. Per-frame chaining (Section 10.3) moves the 32 B link out of each record and into the frame, where it still has to travel. The test had subtracted it from the record and added it nowhere.
- a **record that decodes alone**. 13 B was the *mean delta* record less its link. A frame that verifies without its predecessor starts with a keyframe.

With both charged, the “29 B of room” is $29 - 35 < 0$. DR3 is excluded under either header, whatever the telemetry. And the count had been of seven data rates where the standard defines twelve.

Remark (What the second error teaches that the first did not). The first correction ended with the sentence: *a register that stores facts separately does not compose them; only re-derivation does*. Re-derivation was not enough either. Every number in the first correction was re-derived and every one was right. What was wrong was that no frame of the proposed kind had ever been *built*: the design was a sum of sizes measured in three places. The lean format now exists as code (Chapter 3), the floor above is the length of a frame it emitted, and the question “does it fit” is answered by trying.

The half that was right throughout: DR0–DR2 are excluded by the signature alone.

10.3 Per-frame chaining, and what it is actually worth

On 802.11 freshness binds, so the batch is fixed by $\Lambda_i D_{\max}$ and saving payload bytes buys no extra records. Here the regional payload limit binds, so bytes convert directly into records per frame, and the duty cycle converts records per frame into sustainable rate. Carrying the chain link once per frame instead of once per record is therefore worth far more on this arm.

Remark (Correction, October 2026). Figure 10.2 and the factor of 3.0 first reported with it size every record from a 13 B body — the delta record measured with records 50 ms apart. At the rates this arm sustains, records are seconds apart, and a difference grows with the time it spans:

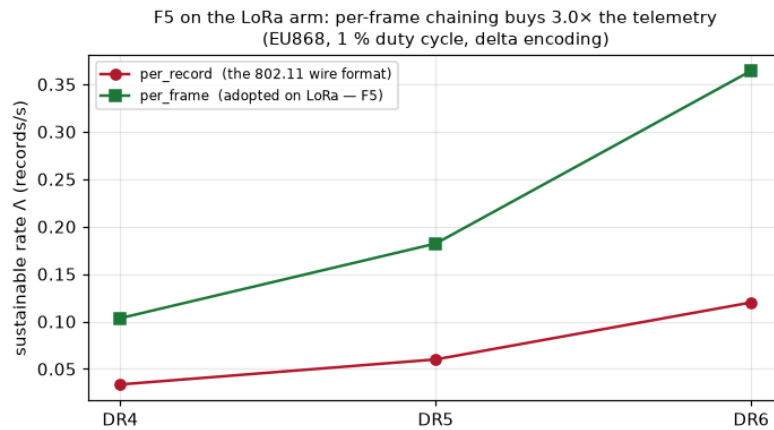


Figure 10.2: Per-frame versus per-record chaining across data rates, as first computed (see the correction below).

spacing	50 ms	0.1 s	1 s	5.5 s	16.5 s
lean delta record (B)	9.0	10.1	12.5	13.6	16.2

The first figures also carried no keyframe, so no frame decoded without the one before it. Sized at the spacing the batch itself implies, with every frame self-contained, DR5 at 242 B gives:

	records per frame	frame (B)	records/s per node
as first reported	7	231	0.182
first format, corrected	5	240	0.13
lean format	8	241	0.20

Per-frame chaining is worth about $2.1\times$ in the first format, not $3.0\times$; the lean format recovers the larger figure because it also shortens the header and the record. The decision to chain per frame on this arm is unaffected: its argument is that the payload limit binds, and that has not changed.

In the first format the 802.11 arm keeps a link in every record, because freshness binds there and the same change buys no extra records. The lean format carries one link per frame on both arms. What that gives up is stated in Chapter 3: inside a frame, order and content rest on the frame signature alone instead of on a signature and an independent hash.

10.4 Capacity

Time on air and the duty cycle are deterministic, so simulating them would be circular. One quantity is not derivable that way — how many nodes can share the channel — be-

cause LoRaWAN uplinks are pure ALOHA, with no carrier sense and no acknowledgement. Simulating LoRaWAN scalability in ns-3 is established practice [36], including proposals to replace the ALOHA MAC with carrier sense [37] and extensions to non-terrestrial links [38].

10.4.1 A topology caveat

LoRaWAN is by specification a star of stars with no peer-to-peer mode: a FANET survey notes that it “presents some limitations regarding its star topology, its medium access control (MAC) layer and its lack of routing procedures” [39], and published peer-to-peer LoRa meshes are built by discarding the LoRaWAN MAC entirely, “not relying on LoRaWAN...without the use of gateways” [40]. A LoRa FANET that was actually flown reports the same constraint, and hybrid LoRa/802.11s UAV meshes reach for a second radio for the same reason [41]; having built one, Zirak *et al.* converge on a single channel and on dedicated transmit and receive radios to work around half-duplex [32], which is the configuration modelled here. A review of mesh proposals finds that even those pursuing flexible topologies keep “a predominant nodes $\rightarrow$ gateway/sink data flow” [42]; concrete multi-hop deployments keep the same sink-directed shape [43, 44], and a 2025 ns-3 LoRa mesh study states that “there is currently no standardised and commercialised multi-hop LoRa-based network” [45].

The scenario is therefore N end devices reporting to one receiver, where the 802.11 arm is a broadcast among peers. What transfers is the part that matters: collision statistics at a receiver depend on how many transmitters are concurrently in range, not on what the receiver is attached to. It is also why a single demodulator is the right provisioning here and not, as it would be for a gateway, an artificially harsh one: a UAV peer has one radio. Two ad hoc effects remain outside the model. A half-duplex radio cannot receive during its own 364 ms frame, costing a peer about 2% of every other peer’s frames. And V is a per-receiver measure, as on the 802.11 arm: a record reaching *every* peer in one broadcast has probability about $(1 - p)^{N-1}$, which is not what V measures and would need gossip or relay.

10.4.2 The result, and how it must be quoted

Simulating the frame at DR5 under the 0.95 threshold gives $N_{\max} = 3$, with a sharp cliff: delivery is 0.960 at $N=3$ and 0.917 at $N=5$. There is no backoff to absorb contention.

Remark (Never quote this bare). The bootstrap interval is $[2, 3]$ — a knife edge — and at the certified $N = 3$, **nine of thirty seeds fail the very criterion being certified**. Under a per-realisation reading, where at least 95% of runs must individually meet V , the answer is **1**, not 3.

Both are computed and emitted. The decision to headline the mean is recorded as

a decision, and it was safe to take only because the comparative result survives either reading.

An earlier value of $N_{\max} = 5$ came from a three-seed sample against a bimodal distribution and is retracted; the artifact that produced it is retained, renamed and superseded.

The capacity runs sent a 218 B frame of six records on the 222 B payload table, which is the table the simulator enforces. The corrected frames on that table are 214.0 B (lean, six records) and 221.7 B (first format, four records), within 2% of it in size and airtime, so the runs were not repeated.

10.4.3 Why DR5 and not DR6

Table 10.1 admits DR4, DR5 and DR6 equally, and DR6 (SF7 at 250 kHz) runs at twice DR5's bit rate, so the choice needs stating. *Capacity does not change.* At a fixed duty cycle d the transmissions falling inside one $2T_{\text{air}}$ vulnerability window number $N \cdot 2T_{\text{air}}/T_{\text{period}} = 2Nd$: **the airtime cancels**, so $P(\text{deliver}) = e^{-2(N-1)d}$ is rate-independent and $N_{\max}$ is the same at either rate. That closed form reproduces the measured DR5 run to within 0.9 percentage points across $N=2-10$ and returns $N_{\max} = 3$, matching the simulation. DR6 therefore doubles the aggregate at the same three nodes, and pays for it in link budget: -120 against -123 dBm is 3 dB, or $1.41\times$ less range under free-space spreading.

We derive this rather than simulate it, and the reason is a limitation worth recording. The module's sensitivity vectors are indexed by *spreading factor alone* — there is no bandwidth dimension in `EndDeviceLoraPhy::SENSITIVITY` or its gateway counterpart, and none in the interference helper. Forcing SF7 onto a 250 kHz channel would have returned DR6's doubled bit rate while silently keeping DR5's noise floor, so the 3 dB penalty — DR6's entire cost — would have been invisible in a run that completed cleanly and repeated across all 30 seeds.

10.4.4 What the configuration models, and what it does not

$N_{\max} = 3$ is far below the node counts usually quoted for LoRaWAN, and the difference is configuration. The simulation uses the LoRa physical layer on its harshest MAC provisioning: a single logical channel, a single demodulation path and one forced spreading factor, since the data rate is a design variable here and each run fixes it. The interference matrix treats *any* same-spreading-factor overlap as fatal, so there is no capture, where a real receiver recovers the stronger of two colliding frames. Enabling a 6 dB capture threshold raises delivery by 2.7 points at $N=8$ and by $1.29\times$ at $N=50$ without moving $N_{\max}$. The small margin over the Poisson curve e^{-2G} at low load is traffic regularity, not capture: sources are periodic, for which the escape probability is $(1 - 2\tau/T)^{N-1} = 0.868$ at $N=8$ against 0.871 measured.

10.4.5 The result is range-limited, and that limit binds first

The channel model attributes all loss to collisions. Enabling correlated shadowing changes the outcome on no row at either 500 or 1000 m, and that is a property of the link budget: with 14 dBm transmit power and -130 dBm SF7 gateway sensitivity, log-distance path loss leaves 28.8 dB of margin at 500 m and 17.5 dB at 1000 m, which a shadowing standard deviation of about 8 dB cannot erode. *The propagation parameters describe a considerably better link than a real drone-to-drone channel.* Field measurements of air-to-air LoRa on real drones [32] show a link losing frames to range alone: delivery falls from 0.9711 at 200 m to 0.9550 at 500 m and 0.9045 at 1000 m, measured with two drones and a base station, where contention is negligible. The two loss mechanisms are independent, so delivery is $P_{\text{link}}(d) \times P_{\text{no-coll}}(N)$:

V target	200 m	500 m	1000 m
0.95	1	1	0
0.90	3	3	1
0.85	5	5	3
0.80	10	8	5

At $V \geq 0.95$ the regime admits no multi-node network at all. The link alone delivers 0.9711 at 200 m, leaving 0.0039 of headroom, while the least-loaded measured point ($N=2$) already spends 0.0283 on collisions. Relaxing to $V \geq 0.90$ restores $N_{\text{max}} = 3$, and to 0.85 restores 5. The curve is reported instead of a point because the choice of threshold, not the physics, decides whether this regime is usable, and that choice belongs to the deployment. **$N_{\text{max}} = 3$ is therefore a worst-case bound at short range — one channel, one demodulator, one spreading factor, $d \leq 500$ m — and not a LoRaWAN network capacity.**

10.4.6 What infrastructure collection would buy

The complementary question — what if a ground gateway collected the ledger — is a configuration change: three channels and eight demodulation paths, with the same forced data rate. Delivery improves substantially and the improvement grows with load, reaching $1.90\times$ at $N=50$ ($0.3755 \rightarrow 0.7134$) and still 0.5071 at $N=100$. **Yet N_{max} moves only from 3 to 8** (95% interval [5, 8]; the bare value must not be quoted). The $V \geq 0.95$ criterion sits on a near-vertical part of the delivery curve — the gateway passes at $N=8$ with 0.9526 and fails at $N=10$ with 0.9398 — so a threefold reduction in offered load per channel moves the threshold barely at all.

10.4.7 External corroboration, quoted as a curve

Haxhibeqiri *et al.* [31] publish a capacity result in closed form — a quintic fitted at $R^2 = 0.997$ to total packet loss measured on SX1301 hardware — so it can be evaluated at this work’s operating point instead of quoted at theirs. It was implemented, checked against the four loss figures their own text states, and given the same criterion.

Table 10.3: The low-rate capacity by two independent methods: frame loss at N transmitters.

N	this work (ns-3)	Haxhibeqiri <i>et al.</i>
2	2.8%	3.1%
3	4.0%	3.8%
5	8.3%	5.1%
8	12.9%	7.0%
30	43.0%	19.9%
50	62.5%	29.9%
$N_{\max}$ at $V \geq 0.95$	3	4

Remark (Two cautions about that comparison). Their $N_{\max}$ is decided at $N = 4$ and $N = 5$, both inside the region their own fit is intercept-dominated: the polynomial predicts 1.78% loss at *zero* transmitters, roughly 40% of the entire loss budget the criterion allows. Forcing the fit through the origin would give them $N_{\max} = 5$, *widening* the gap — so quoting 4 is the conservative choice.

And the pessimism ratio is not a single number. It runs $0.91\times$ at $N = 2$, where *this work is the more optimistic model*, through $1.07\times$ at $N = 3$, to $2.1\times$ from $N = 10$ upward. The sign changes in exactly the region where $N_{\max}$ is decided. An earlier finding in this project quoted a single ratio in the wrong direction and had to be retracted.

Remark (A correction of attribution). Until October 2026 this model was cited throughout the project — in this chapter, the paper, the code and eight documents — under the names of the authors of a *different* paper. The source had been read and its figures checked; nobody had compared the author list with the title page. Every bibliography entry is now compared with its registry record by a script (Chapter 11).

10.5 What the regime is for

Three nodes at about 0.17 records per second each are 0.5 records per second in all, where the 802.11 design carries some 6200. The low-rate link is not a slow 802.11. Its defensible role for a ledger is long-range *provenance* — a tamper-evident trickle — and not live telemetry. An independent ns-3 comparison of the two technologies reaches the same structural conclusion from a different direction: Klimiashvili *et al.* find neither uniformly better and recommend selecting between them instead of tuning one [34].

10.6 Scope of this chapter

This is simulation and analysis, not measurement. No radio was metered, and there is no energy column. Airtime per record is reported instead: it is what the regulator rations, and it is computed, not assumed.

Chapter 11

Reproducibility, Defects and Research Integrity

► *STATUS: DRAFTED. This chapter is unusual for a thesis and deliberate. It reports the errors found in this project's own results. Examiners should read it as the methodological contribution it is intended to be, not as an admission appended to a limitations section.*

11.1 Why this chapter exists

Most theses report what was found. This one also reports *what was found to be wrong, and how*. The justification is not confession but transferability: the defects catalogued below are not peculiar to this work, and several of them survived safeguards that are widely considered sufficient — version control, a green test suite, seeded runs, committed raw data.

By August 2026 ten results in this project had moved after they were first recorded. Only four were catchable by running more seeds. Two were *protected by passing tests*. Three were contradictions between a paper and the artifact that generated it.

In October an external review of the paper found five more, and they were of a kind the first ten had not prepared the project for (Section 11.3). Every one of them had survived two full audits and a green suite of thirteen hundred tests, and one of them was a “correction” the second audit had itself introduced.

11.2 The defect taxonomy

class	mechanism	instance in this work
C1	small-sample mean lands the wrong side of a threshold	four headline numbers; a capacity value $5 \rightarrow 3$
C2	an unverified constant on the measurement path	the 2.4 GHz basic-rate saturation; the frame header treated as a constant
C3	a threshold applied to a mean rather than a distribution	the capacity criterion; the utilisation crossing
C4	a configuration change that perturbs the random realisation	simulator stream displacement under mobility
C5	a claim wider than the experiment supports	two-node hardware described as validating contention; a load ceiling measured at one N applied at all
C6	a correction made once, at its source, that does not reach the copies	five constants derived from a re-measured header; a retracted finding still printed in a table
C7	a composite that was never built: a result summed from parts measured separately	the headline design; its verifiability; the exclusion test; the low-rate batch
C8	a simulated source more regular than any real one	phase-locked senders, on both arms
C9	a source taken on trust	a paper cited for two months under another paper's authors; figures quoted from a paper that was not held

Remark (The load-bearing observation). **More seeds fixes only C1.** C2–C9 are immune to sample size. “We ran thirty seeds” is therefore not a statement of correctness, and the discipline installed after the C1 failures gave no protection at all against the C2 failures found later — nor did the discipline installed after those protect against C7.

11.3 October 2026: five errors that survived two audits

The design had never been built. The configuration the results headlined — delta records, one signature per four — was a *sum of sizes*: a record size from one module, a header measured on frames of a different encoding from another, and a signature

length. No frame carrying delta records had been encoded, sent or decoded, and the receiver that would decode one did not exist. Each number was individually correct, so re-deriving any one of them reproduced it.

Its verifiability was wrong. The byte model amortised one keyframe over four frames; the verifiability model treated the four as independent. At its own loss budget the design verified 0.881 of its records, not 0.95 (Theorem 4.6).

The exclusion test left out two things a frame must carry — the chain link and a record that decodes alone — and counted seven data rates where the standard defines twelve. The August audit had “corrected” the exclusion from four rates to three on the strength of a leaner header; with the link charged, that correction is itself wrong (Section 10.2.5).

A record size was used at the wrong time scale. The delta record measured with records 50 ms apart was used on the low-rate arm, where they are seconds apart and the same record is half as large again (Section 10.3).

A source was cited under the wrong authors, in the paper, the thesis, the code and eight documents, for two months. Each citation had been “verified”: the DOI resolved and the claim was in the PDF. Nobody had compared the author list with the title page — and the paper carried a paragraph asserting that every citation had been checked.

The first four are one error seen four times. What fixed them was not a better audit but an object: the frame format was implemented, with a sender and a receiver, and the byte, loss and exclusion results were recomputed *from one definition of a frame*, whose arithmetic a test holds equal to the frames the code emits. The fifth was fixed by a script that compares every bibliography entry with its registry record, and by deleting the paragraph: a claim of that kind belongs in a check that runs, not in prose.

11.3.1 The capacity ceiling, and a prediction that failed

The same review asked for confidence intervals on the 802.11 capacities. Producing them meant simulating each configuration directly instead of applying one load ceiling, and the prediction that the two would agree within $\pm 10\%$ — committed before the runs — failed (Chapter 9). The ratios this thesis had described as “protected by construction” moved with it.

Chasing the per-run spread of those simulations produced the 802.11 counterpart of an artifact already known on the low-rate arm (below): strictly periodic senders keep their relative phases for a whole run, so a run is one phase configuration, and in an unlucky one a few nodes lose every frame while the rest lose none. Three follow-up experiments were registered before they were run. Two of their predictions failed and are reported as

failures; the third, that the choice of source moves the mean by less than a tolerance set in advance, held; with all six cells run, the difference is small, consistently of one sign, and not yet explained (Section 8.5).

11.4 A simulator artifact large enough to change what a reader should believe

The ns-3 LoRaWAN module’s periodic sender fires on an exact interval, and LoRaWAN ALOHA has no backoff, so relative transmission phases are frozen for an entire run: a pair that collides once collides on every cycle. Delivery becomes *bimodal* instead of noisy about a mean. Over 30 seeds at $N=5$, 22 runs deliver exactly 1.000 and the rest fall to 0.20–0.84. The coefficient of variation is inflated **2–8×** relative to a sender with one-second transmission jitter, the inflation growing with node count ($2.9\times$ at $N=5$, $7.9\times$ at $N=100$). The LoRaWAN specification makes it a rule that an end-device “pseudo-randomly changes its transmit periodicity to prevent systematic synchronization of populations of end-device transmissions” [18]; the module’s sender does not.

The measurement is claimed, not the observation. Durand and Booyen report the same effect in their own results, attributing bimodal delivery to nodes that “always transmit on a specific SF, time, and channel”, producing “certain packet collisions being repeated for every transmission cycle” [45]. What is added here is the quantification.

On 802.11 carrier sense softens the effect and does not remove it. The simulator declares the medium busy only at the end of a $4\mu\text{s}$ preamble-detection period; two strictly periodic nodes whose phases fall inside that window collide in every period. At $N = 29$ the per-run standard deviation of delivery is 0.034 with strictly periodic sources and 0.0016 when phases are redrawn every period, with a mean that differs by 0.003 over the points run under both.

11.4.1 A claim that was withdrawn

It had been intended to report that this dispersion is undetectable in the ns-3 LoRa literature because that literature does not state replication information, and a survey was pre-registered to test it — inclusion criteria, keyword set and a 25% falsification threshold, all committed before the corpus was assembled. At the achieved $n=23$ the result is 21.7% reporting with a 95% interval of [7.5, 43.7]%. **That interval contains the threshold, so the survey cannot answer its own question, and no claim about the literature is made.** The protocol, corpus and adjudications are in the repository as a null result.

11.5 An automated staleness gate, and what it does not cover

Every model-derived artifact is re-derived from current code on every run and compared row by row; a single differing value fails the build. This makes silent staleness of *derived* data impossible.

It does not make staleness impossible. Two limits were found the hard way:

- **Scope.** Every drift guard originally parsed the paper. The specification documents were outside that scope, so corrections propagated into the paper and not into the documents for three weeks — including a table built verbatim on data the project had already purged and superseded.
- **Direction.** Retracting a finding in a register does not remove it from the prose. One retracted claim was found alive in the paper months later, one hundred lines below a sentence stating the opposite, and later again in the open-items register. **When you retract, grep the whole repository.**

11.6 Practices that earned their place

Pre-registration, committed data-free. Predictions are committed as their own commit before data exist, so the ordering is third-party checkable. This cost the project a claim once: an assertion of pre-registration was *withdrawn* rather than reconstructed, because writing an expectations file after results are known manufactures evidence.

Stating the expected value before measuring. This is what caught the 2.4 GHz basic-rate artifact, which repeatability would never have caught.

Retaining a discarded model as a labelled failure curve. The naive broadcast reduction remains in the codebase, clearly marked, so the 16× error it produces can be shown rather than described.

Keeping per-seed raw data, not summaries. This allowed validation bands to be re-derived at thirty seeds without re-simulating.

An independent second implementation. The slot-exact simulator, written before the published model was found, shares none of its assumptions and is what turns agreement into evidence.

No typed results. Since October 2026 every result in the paper, and every corrected result in this thesis, is a macro in a file that a script writes from the artifacts. A sentence that states a verdict instead of a number — “eight of twelve”, “does not

meet the target” — is held by the same script, which refuses to run if the artifact no longer says so.

A check on the bibliography that runs. Every entry is compared with its Crossref, DataCite or arXiv record — family names in order, initials, title, year — and entries with no registry record must name the held document they were checked against.

11.7 Two defects worth stating individually

11.7.1 A test that asserted a bug

A utilisation function returned exactly zero for a single node, with a unit test asserting that a lone sender never contends. The function conflated *contention* with *utilisation*: a lone sender does not collide but still occupies airtime, and returning zero declared a single node able to carry unbounded traffic. The defect was harmless until another result began reporting single-node cases — **and it was protected by a passing test that had encoded the wrong behaviour as intended behaviour.**

11.7.2 A criterion that could not fail

The project pre-registered a success criterion requiring at least a 40% reduction in authentication bytes, and committed it to version control two days before the result existed. The ordering is genuine and verifiable.

The quantity it tests, however, reduces exactly to $1 - 1/b$. The threshold is therefore $b \geq 2$ — independent of encoding, scheme, placement, header size and signature size. The companion condition on verifiability is satisfied by construction with zero margin. **No configuration that batches at all could have failed the test.**

The pre-registration is real and worth keeping; what had to go was the implication that it constituted a risk the design might not have survived.

11.8 What a reader should take from this

Three times before October 2026 a claim in this project survived because two facts were each individually recorded and never placed side by side, and this chapter used to end: a register that stores facts separately does not compose them; *only re-derivation does*.

That was not enough. The errors of Section 11.3 were re-derived, repeatedly, and reproduced every time, because each part was right. **What composes facts is building the thing they describe and making it run.** A design that exists only as a sum has no receiver to refuse a frame it cannot decode, and no encoder to refuse a frame that does not fit.

► *TO ADD: a comparison with the reproducibility literature — the simulation-credibility lineage in particular — so this chapter reads as a contribution to that conversation rather than as a project post-mortem. The companion methods paper does this; the material needs porting.*

Chapter 12

Conclusions, Limitations and Future Work

► *STATUS: DRAFTED. The future-work section is the one most likely to be examined for evidence that the candidate understands the work’s boundaries; it is written accordingly and deliberately does not oversell.*

12.1 What this thesis establishes

1. After compression, authentication — not data — is the dominant on-air cost of a tamper-evident UAV telemetry ledger. At the smallest encoding studied, the signature is larger than the record it signs.
2. Frame-level self-batching is the correct placement, and it Pareto-dominates block-level aggregation under *any* stationary loss process, not merely under the independent-loss assumption the emulator implements.
3. On IEEE 802.11 the freshness deadline binds before the frame size does, so the amplification law that motivates aggressive compression is never operative: compression pays exactly $1\times$. On low-rate links the frame size binds and the leverage is real. **The regimes are different in kind.**
4. A delta-coded frame that depends on its predecessor does not meet a verifiability target equal to the loss rate. Every frame of the design therefore carries a self-contained record, and the cost of that is small.
5. Eight of the twelve EU863-870 data rates cannot carry one signed, hash-chained frame: five because a 64 B signature does not fit a 51 B payload, three because header, chain link and signature fill a 115 B payload. The result holds under four stated conditions.

6. Built as one frame format and simulated directly, the design carries 43.25 B per record where per-record signing carries 145.77 B (70.3% fewer), and sustains 124 neighbours at 95% delivery where per-record signing sustains 35, at an operating point compliant with 3GPP TS 22.125.
7. Of the four design axes, placement and batching interact exactly, by $g_a(1 - 1/b)$, and in the frame as built delta coding is conditional on batching as well: both gains are zero at $b = 1$. The scheme axis is byte-neutral. The four-axis co-design claim, as originally stated, was overstated; what is true is that batching is the axis the other gains depend on.

12.2 Limitations, stated as such

12.2.1 Modelling scope

The single collision domain admits no spatial reuse and no hidden terminals; a real formation has both, one helping and one hurting. Mobility was tested and found null for collision-limited capacity, but per-frame Doppler is unmodelled — the null result means mobility does not change *collision-limited capacity*, not that mobility is harmless to a link. Loss is independent per frame; the ordering result survives correlation but absolute verifiability figures do not. The keyframe interval is fixed and unoptimised. Post-quantum sizes are projections, not implementations.

12.2.2 Validation scope

Hardware validates *airtime* on a two-node link; contention is simulated only, and the capacities of Chapter 8 rest on that simulation. Record sizes come from a synthetic generator, checked against twelve public flight logs at 0.2 s spacing and above and against nothing at the 50 Hz of the operating point. Only the sender's energy is metered, and only for the first format; composed energy is a lower bound by roughly 7–14%, the residual being uncharged prototype framing overhead — measured, stated, not corrected away. Capacity was simulated at one PHY rate.

12.2.3 The low-rate arm

Analysis and simulation only: no radio metered, no energy column, one spreading factor per run. The capacity figure is a within-one-spreading-factor bound, not a network capacity, and its confidence interval is a knife edge.

12.2.4 The boundary itself

The exclusion is conditional: on LoRaWAN's regional payload limits, on a public-key signature in every frame, on no loss recovery, and on a verifiability target no looser than the loss rate. It was also stated wrongly twice before reaching its present form (Section 10.2.5), and a reader is entitled to weigh that. What has not moved in any version is the part that rests on one inequality: $64 > 51$.

12.2.5 Security scope

Confidentiality, availability, key management and protection against an attacker who delays frames are outside the design (Section 3.4). With one chain link per frame, continuity inside a frame rests on the frame signature alone.

12.3 Future work

12.3.1 Immediately available, and cheap

Pricing implicit certificates. The certificate accounting charges an explicit certificate periodically. Implicit (ECQV) certificates are the standard smaller alternative and were never modelled. The current charge is an upper bound, which cuts against this work's own comparison — so the gap is safe but should be closed.

Timing the lean codec on the target board. Its sizes are measured; its encode and decode times on the Raspberry Pi are not, so the energy table covers the first format only.

Telemetry at the operating rate. Public flight logs carry position at 5 Hz. A log recorded at 50 Hz would settle the one size — the delta record at 20 ms spacing — that currently rests on the generator alone.

12.3.2 Substantial, and worth doing

Channel-access delay in the freshness model. The freshness constraint that *sets* the batch omits the dominant delay term at high load. It is currently mitigated by reporting utilisation alongside, not by fixing the model. A DCF access-delay term would make the freshness constraint credible at high utilisation, where it currently is not.

Multi-spreading-factor capacity on the low-rate arm. Each run fixes one data rate, forfeiting the quasi-orthogonality that supplies much of a real deployment's capacity. That answers a different question — deployment capacity rather than per-configuration feasibility — and it is the question an operator would ask next.

Contention on real hardware. Two radios validate airtime; the capacities are simulated. Three or four radios in one collision domain would not reach the crossing, but would test the one thing the simulation’s capacities depend on that no measurement here touches: loss under contention at a known offered load. It is the single most valuable missing measurement in this thesis.

A post-quantum arm that is implemented rather than projected. The projections show that batching cannot rescue post-quantum signature sizes, because freshness caps the batch and one signature plus header already exceeds the frame budget. That is a strong negative result and it deserves measurement rather than arithmetic.

12.3.3 Speculative, and stated as such

The exclusion bound is link-agnostic. Instantiating it on other constrained bands — and on the signature sizes the post-quantum standards actually mandate — would say which of tomorrow’s radio and cryptographic combinations are feasible before either is deployed. That is the natural generalisation, and it is the one this author would pursue.

12.4 Closing

This thesis introduces no new cryptographic primitive and makes no claim to one. Its contribution is a reproducible co-design of existing, standardised components; the quantified trade-offs that co-design entails; the boundary beyond which no batch size helps, with its conditions; and an account — unusually explicit for a thesis — of the errors made in establishing that boundary and the classes of defect that survived the safeguards intended to prevent them.

The most durable result here is also the smallest one that survived scrutiny: on five EU863-870 data rates, a 64-byte signature does not fit a 51-byte payload, and no batch size, encoding or placement changes that. A shorter authenticator does, and Chapter 10 says what it buys.

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